

Hours and Wages in Equilibrium^{*}

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Abstract

There are substantial differences in the long-hour wage premium across occupations, which are important to understand gendered sorting in the labor market. To explain why these different premiums emerge, previous work emphasizes the importance of heterogeneity in how hours enter the firm's production function. But hours and wages are equilibrium outcomes, which also depend on the market structure. I build a tractable directed search and matching framework with two-sided multidimensional heterogeneity, in which hours and wages are jointly determined as a function of workers' preferences, firms' production technologies, and tightness in the labor market. I use the model to estimate the occupational long hour premiums in the United States, and decompose them into a productivity and bargaining component. These respectively explain about 82% and 18% of the premium. A counterfactual analysis shows that halving the productivity premium associated with long weekly hours reduces the gender wage gap by a moderate 5%.

Preliminary and Incomplete – Please do not Circulate

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1 Introduction

There is substantial variation in hours and the wage premium associated with long weekly hours across occupations. A seminal paper by [Goldin \(2014\)](#) argues that these differences are key to understanding gendered sorting patterns in the labor market, and contribute significantly to inequality at the upper end of the earnings distribution. To explain why long-hour premiums emerge, [Goldin \(2014\)](#) and [Goldin and Katz \(2016\)](#) emphasize the importance of differences in how hours enter the production function. But hours and wages are equilibrium outcomes, which also depend on the preferences of workers and the market structure. I empirically document an important role for labor market tightness in shaping the equilibrium distributions of both hours and wages. Then, I introduce a tractable equilibrium search and matching framework with endogenous hours and wages to explain how long-hour premiums are shaped.

When markets are tighter, workers have stronger bargaining positions, which they can leverage to negotiate better wages, hours, or extract larger long-hour premiums. I empirically assess whether these margins adjust using data from the latest [American Community Survey \(ACS\)](#), merged with occupational vacancy distributions constructed from the [National Labor Exchange \(NLx\)](#) that are scaled through [Job Openings and Labor Turnover Survey \(JOLTS\)](#). These allow me to introduce a measure of market tightness that is particular to a given worker and occupation type. In the high-end labor market of [Goldin \(2014\)](#), a percentage increase in the vacancy to unemployment ratio raises average wages by 0.02% and reduces hours by up to 0.03%. Moreover, the wages of individuals who work longer hours increase disproportionately, as a percentage increase in tightness raises the returns to hours worked by about 0.01%.

To interpret these patterns, I introduce a model of search and matching in the labor market, in which agents direct their search efforts as in [Moen \(1997\)](#), but with two-sided multidimensional heterogeneity as in [Choo and Siow \(2006\)](#). Firms post vacancies that specify the wages they are willing to pay for different types of workers at various hours arrangements. Their postings reflect the productive value of a worker at the given number of hours, but also account for the probability of filling the vacancy. Workers observe all offers and simultaneously apply to their favored positions. They take into account a similar trade-off between their preferences over wages and amenities compared to the probability of getting hired. The model yields rich

equilibrium distributions of hours and wages that reflect the production technologies of firms, the preferences of workers, and tightness in the labor market.

I build on the approach of [Choo and Siow \(2006\)](#) and [Galichon and Salanié \(2022\)](#) for estimation. They introduce an empirically tractable framework for estimating frictionless equilibrium models with perfectly transferable utility (TU) that allows for two-sided multidimensional heterogeneity. Introducing search frictions into their model has two important implications. On the one hand, the model better rationalizes the coexistence of unemployed workers and vacant jobs, which are the consequence of coordination failures. On the other, because agents commit to the market they choose to search in, the matching problem has an imperfectly transferable utility (ITU) structure, as in [Legros and Newman \(2007\)](#) and [Galichon et al. \(2019\)](#). This implies that equilibrium hours and long-hour premiums respond to changes in market tightness, unlike in TU models where they are pinned down by preferences and production technologies only.

To estimate the model, I leverage cross-sectional data on matching patterns and wages from the United States, and focus again on the high-end labor market of [Goldin \(2014\)](#). The first main result confirms the narrative of [Goldin \(2014\)](#) and [Goldin and Katz \(2016\)](#) about the importance of production function heterogeneity. The productivity premium to working long hours is substantial in all Business occupations, but less so in Health, Tech, Science or Others. In the highest earning Business occupations, working long hours yields a monthly productivity premium exceeding 10,000 dollars. This translates into substantial equilibrium wage premiums of up to 8,000 dollars per month. The occupations of Medical Practitioners and Computer programmers are characterized by the largest productivity premiums to long working hours outside of Business occupations. At the other end, for Clinical Professionals, Post-Secondary Teachers, and Administration and Training, the additional value created by long weekly hours is smaller, at less than 2,000 dollars per month.

The second main result is that bargaining power differs significantly across occupations, which has important implications for wages but also wage premiums. When the number of occupational vacancies is large, we see that firms have less bargaining power, inflating wages. This is particularly the case for Clinical Professionals, Managers, Business Operations, Sales, and all Tech occupations. But wages are not affected uniformly. Even within the same occupation, bargaining power can differ between the full-time and long-hour markets, which can inflate or

reduce the long-hour wage premium. To better understand how much this matters, I decompose the equilibrium long-hour wage premium into two components. The first solely reflects differences in the productive value of working full time as opposed to long hours, while the second captures the difference in how much firms can capture of the joint surplus, which is determined by tightness in the market. About 82% of the wage premium is due to productive differences, while the remaining 18% stems from differences in the bargained surplus. There is again substantial heterogeneity across occupations in both the sign and magnitude of these effects.

Finally, I study the effectiveness of policies aimed at reducing the large long-hour wage premium. As shown in [Goldin and Katz \(2016\)](#), the productivity and wage premium associated with long hours of work can change significantly over time, even within occupations. To study the effect of these policies, we study a counterfactual scenario in which the productivity premium for working long hours is halved across occupations. This pushes down the earnings of men, but also of women in long hour jobs. The effect on the gender wage gap is moderate is a moderate reduction of about 5%. This means that alternative policies may be more suited if the aim is gender equality in earnings.

Outline. The remainder of this paper is structured as follows. Section 2 discusses the literature that studies the relation between hours and wages, and compares our framework to related models of matching with and without search frictions. Section 3 introduces the data and presents empirical patterns relating hours, wages, and market tightness. Section 4 introduces a tractable equilibrium search and matching model to further interpret the empirical patterns. Section 5 discusses identification and structural estimation of the model. Section 6 presents the estimated parameters and studies how equilibrium hours and wages are shaped. Section 7 concludes.

2 Related Literature

Hours and Wages. This paper contributes to the large literature that studies the relation between hours and wages. A substantial body of work studies non-linearities at the bottom, highlighting large penalties for part-time work ([Macpherson and Hirsch, 1995](#); [Aaronson and French, 2004](#); [Manning and Petrongolo, 2008](#)). More recently, several papers document non-linearities

at the top, finding large returns to long weekly hours (Goldin, 2014; Cortés and Pan, 2019; Mantovani, 2023). There is substantial heterogeneity in these returns, which are mainly prevalent in high-earning occupations. Recent work by Yurdagul (2017) and Bick et al. (2022) shows that the aggregate pattern, including both high- and low-earning occupations, is better described by a hump shape.

Goldin (2014) and Goldin and Katz (2016) provide micro-foundations for the equilibrium long-hour premiums. They argue that the convexity of firms' production functions in hours differs substantially across occupations. Recent papers by Erosa et al. (2022), Jang and Yum (2022), and Erosa et al. (2024) embed these non-linearities into joint models of labor supply and occupation choice. They study the sorting of workers who choose between two occupations that offer different premiums for working long hours. We instead build introduce these non-linearities in an equilibrium model with endogenous hours and wages, and allow for rich heterogeneity on both sides of the market. Contrary to previous papers, we impose prior restrictions about how hours enter an occupation's production function, and separately identify the shape of the production functions from the equilibrium wage function.

Separable Matching Models. We estimate our model by building on the seminal paper of Choo and Siow (2006), who embed McFadden (1978)'s random utility model into a frictionless two sided matching framework. Their approach to estimating equilibrium matching models allows for rich two sided heterogeneity, and has become the benchmark to empirically study marital sorting.¹ But so far, labor market applications remain limited, with exceptions that mainly focus on low friction markets (Dupuy et al., 2020; Dupuy and Galichon, 2022; Corblet, 2023; Dupuy et al., 2023). An important reason is that the model assumes away search frictions, which is at odds with the large literature that followed Diamond (1982), Mortensen (1982), and Pissarides (1985).

We add search frictions into a separable matching model, and thus relate to Jaffe and Weber (2019), Ciscato (2024), and San (2024). They introduce heterogeneous meeting costs between agents of different types into models based on Choo and Siow (2006). An important difference is

¹Some notable recent examples of studies that use (extensions of) their approach to study marriage markets are Dupuy and Galichon (2014), Choo (2015), Siow (2015), Greenwood et al. (2016), Chiappori et al. (2017), Chiappori et al. (2018), Ciscato and Weber (2020), Chiappori et al. (2020), Galichon and Salanié (2022), Ciscato (2024), Reynoso (2024), and Chiappori et al. (2025).

that we assume agents search and match in local markets, as in standard directed search models (Moen, 1997; Chade et al., 2017). We thus impose no centralized market clearing condition, since agents fully commit to their chosen local market. This has nontrivial implications in terms of utility being only imperfectly transferable (ITU) between workers and firms (Legros and Newman, 2007; Galichon et al., 2019). By relaxing the usual (perfectly) transferable utility (TU) assumption, we allow for market tightness to affect the endogenous matching terms. Similar extensions have recently been introduced in Mourifié (2019) and Mourifié and Siow (2021), but without a clear structural underpinning.

The models in Arcidiacono et al. (2016) and Beauchamp et al. (2024) are also closely related, since they also consider directed search models with two sided multidimensional heterogeneity that build on Choo and Siow (2006). But they study the marriage market, where information on transfers is not readily available. To identify preferences on both sides of the market, they assume that utility cannot be transferred between potential partners (NTU). While this can be justified in particular settings, it is a strong assumption in modern marriage or labor markets, because it rules out any concession between workers and firms. We relax this by allowing for utility to be imperfectly transferable.

Search Models with Amenities. We also add to the literature that considers the identification and estimation of search models where workers value both wages and amenities (Lavetti, 2023; Mas, 2025). As noted in Goldin (2014), the long-hour premium essentially reflects a compensating differential for the disamenity of working long weekly hours. Several papers have introduced frictions into hedonic models à la Rosen (1986), aiming to describe the equilibrium relation between wages and amenities. Contrary to the competitive Rosen (1986) benchmark, in Hwang et al. (1998) and Lang and Majumdar (2004) search frictions may prevent wages from reflecting the underlying preferences. Bonhomme and Jolivet (2009) estimate a partial equilibrium version of their model, and find relatively large compensating differentials for working times, among other amenities. Flabbi and Moro (2012) estimate a similar search model with bargaining, and find that up to one third of women would sacrifice wages for the flexibility of working part-time. While our framework is similar in terms of equilibrium wages not necessarily reflecting preferences due to search frictions, we estimate the equilibrium model, and allow workers to direct their search

efforts as in [Peters \(1984\)](#), [Montgomery \(1991\)](#), or [Moen \(1997\)](#). This introduces segmentation into the market, which allows for market tightness to affect both hours and wages.

Alternatively, several papers study the share of wage variation that can be explained by amenities using partial or general equilibrium models with search frictions. To do so, they typically rely on job-to-job transitions and revealed preference arguments ([Sullivan and To, 2014](#); [Sorkin, 2018](#); [Taber and Vejlín, 2020](#)). This literature highlights the importance of compensating differentials, which explain between 15 and 30% of earnings inequality. We instead focus on the compensating differential for a particular amenity, and identify the model from cross-sectional variation in matching patterns and wages in several markets.

3 Empirical Patterns

This section introduces the data we use to analyze the relation between hours, earnings, and tightness in the labor market. We first classify individual workers and occupations into discrete types, and highlight substantial variation in hours and wages across these types. Next, we show that there are large differences in long-hour premiums across occupations. Finally, we point out disparities in the occupational distributions of matches and vacancies, and estimate how hours and wages respond to changes in the vacancy-to-unemployment ratio.

Data. The primary data source is the latest wave of the [American Community Survey \(ACS, 2023\)](#), provided by the [Integrated Public Use Microdata Series \(IPUMS\)](#). We use the ACS to construct a sample of the high-end labor force in the United States. As in [Goldin \(2014\)](#), we select only individuals between 25 and 64 years old who have at least a bachelor’s degree and work or used to work in a high earning occupation as listed in Table 1. To construct an employed sub-sample, we select only those that work full-time.² We also construct an unemployed sub-sample, selecting those whose last job was in a high earning occupation but who are currently unemployed. This assures that the worker-type distribution among the unemployed is representative for the high-end market.

²We define full-time work as at least 35 hours per week. Because we focus on college educated workers in high-earning occupations, there is little part-time work. Approximately 11% of individuals reports working less than 35 hours per week or 49 weeks per year.

Table 1: Types of Occupations

Business	Health	Tech	Science	Other
Lawyers and Judges	Medical Practitioners	Comp. Programmers	Natural Scientists	Post-Sec. Teachers
Exec's and GMs	Adv. Clin. Prof.	Engineers	Social Scientists	Public Safety
Managers	Clin. Prof.	Cybersec. & Infrastr.		Other
Financial Ops				
Business Ops				
Sales				
Admin. & Training				

Notes. Classification of occupations into types similar to those introduced in [Goldin \(2014\)](#). The individual occupations within each type can be found in Appendix [B.1](#).

We then merge in job vacancies from the [National Labor Exchange \(NLx\)](#) Research Hub. Their dataset consists of individual postings collected from companies' career websites, state job banks, and the federal jobs portal. They attempt to cover all online vacancies in the United States, and represent roughly half of all openings as estimated in the [Job Openings and Labor Turnover Survey \(JOLTS\)](#) by the [Bureau of Labor Statistics \(BLS\)](#). We use the NLx data to construct the occupational vacancy distributions, and scale them through JOLTS. More information can be found in Appendix [B.2](#).

Constructing Types. We group workers, firms, and hours into discrete multidimensional types based on observable characteristics. We define $X = 8$ worker types that classify individuals according to their gender, age, and education level. The age dimension separates workers that are older or younger than forty. The education level makes a distinction between those with a Bachelor's, and those with at least a Master's degree. We also define $Y = 18$ occupation types. These follow the broad classification of [Goldin \(2014\)](#), separating Business, Health, Tech, Science, and Other occupations. The full classification is shown in Table 1, and Appendix [B.1](#) lists the individual occupations within each type. Finally, we also define $Z = 2$ hours types, which classify workers into groups based on whether they work full-time (35 to 48) or long (more than 48) weekly hours. We thus have 288 sub-markets that are observed across $S = 9$ independent regions.³

³We aggregate states into Census Divisions as defined by the [Census Bureau](#) to be consistent with the model.

3.1 Hours and Earnings

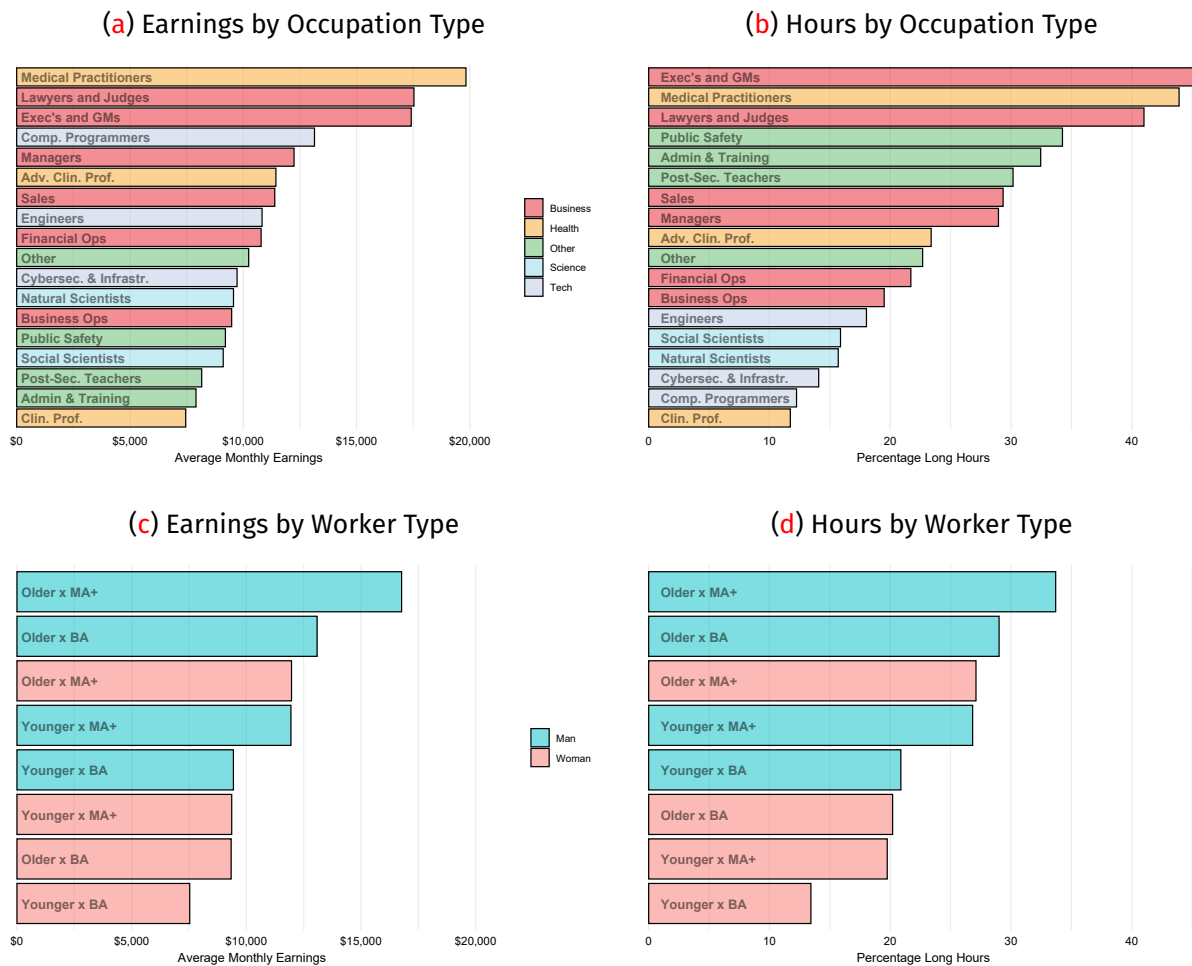
Occupation Types. We first look at variation in hours and monthly earnings across the different worker and occupation types in Figure 1. Both hours and earnings are Winsorized at the 1st and 99th percentiles. Panel (a) highlights earnings differences across occupations. The highest earning groups in our sample are Medical Practitioners, Executives, Lawyers, and Judges, who all earn between 17.000 and 20.000 dollars per month. Just below them, we find upper-level Business occupations (Managers, Sales, and Financial Operations), Computer Programmers, Engineers, and Advanced Clinical Professionals, which all earn 10.000 to 13.000 dollars per month. At the lower end, Clinical Professionals, and upper level Administration, Training, and Teaching positions, have average monthly salaries between 7.500 and 8.500 dollars.

Panel (b) highlights differences in hours across the same groups. On the upper end of the hours distribution, we find the highest earning occupations of Executives, Lawyers, Judges, and Medical Practitioners. Among them, 40 to 45 percent works long weekly hours. On the lower end, we generally find the lower earning occupations, such as Clinical Professionals and Scientists. Among them, 15 percent or less reports working long weekly hours. While we generally see that higher earning occupations are characterized by longer hours, there are important exceptions. Particularly, Tech occupations such as Computer Programmers and Engineers are characterized by both high earnings and a low incidence of long working hours.

Worker Types. We now look at earnings differences across worker types in Panel (c). This shows that older men are the highest paid group in our sample, irrespective of education. Their average monthly earnings are between 12.500 and almost 17.500 dollars for those with at least a Master's degree. The next best paid are older women and younger men with at least a Master's, both groups earning close to 12.000 dollars per month. The remaining groups earn less, averaging between 7.500 and 9.000 dollars per month. These earnings differences are well known, see for example the review by [Blau and Kahn \(2017\)](#) that covers both gender differences in earnings and how these relate to education.

Finally, Panel (d) highlights differences in hours across worker types. The first main pattern is that, within each type, men are more likely to work long hours than women. This difference

Figure 1: Earnings and Hours by Type



Notes. Earnings and hours by worker and occupation type in the United States. Data from the [American Community Survey](#) (ACS, wave 2023). Panels (a) and (c) show the distribution of monthly earnings across the occupation and worker types defined in section 3. Panels (b) and (d) show the fraction of workers that work long hours, defined as more than 48 per week, across the same types.

is particularly pronounced for older workers, among whom almost 30 percent of men with a Bachelor's degree work long hours, increasing to almost 35 percent among men with at least a Master's degree. The corresponding percentages among women of the same type are closer to 20 and 25 percent respectively. The second main pattern is that younger women are very unlikely to work long hours. Particularly, if they obtained only a Bachelor's degree, less than 15 percent works long hours. These patterns are in line with the narrative of [Goldin \(2014\)](#) that relates long hours among men to the gender gap in earnings.

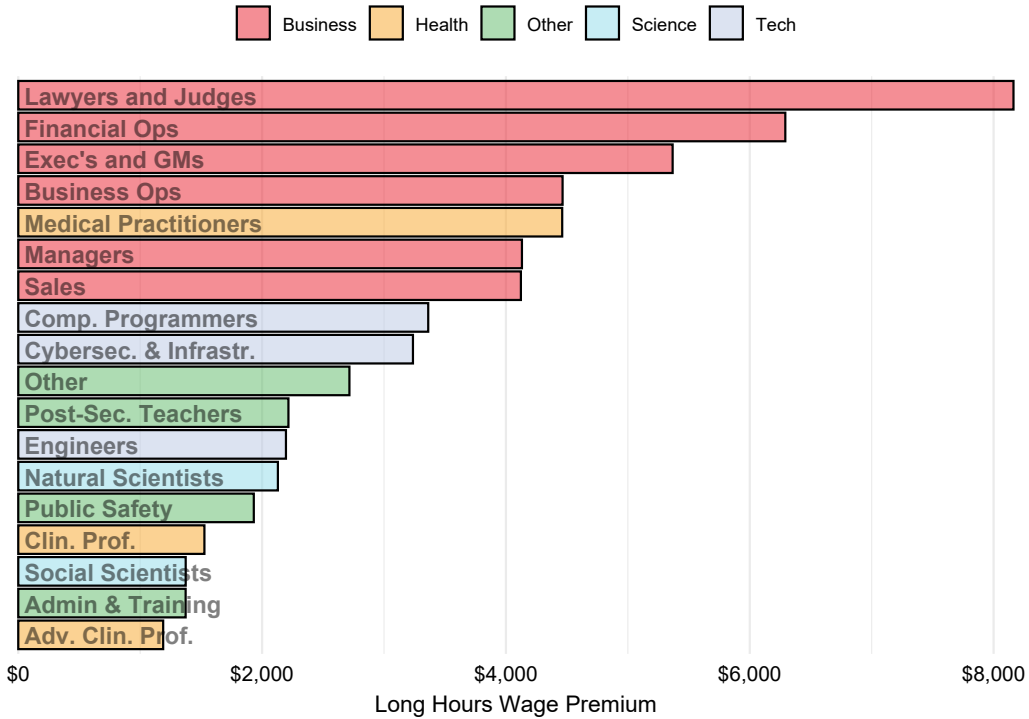
3.2 Long-Hour Premiums

To better understand the equilibrium relation between hours and earnings, we study how the long-hour premium varies across occupations. We do so by estimating regressions similar to those in [Goldin \(2014\)](#) and [Cortés and Pan \(2019\)](#). For each occupation type $y \in Y$ we estimate:

$$W_i = \alpha + \beta \mathbb{1}_{z_i} + \sum_x \mathbb{1}_{x_i} \gamma_x + \sum_s \mathbb{1}_{s_i} \gamma_s + \varepsilon_i, \quad (3.1)$$

where W_i is the wage of individual i and $\mathbb{1}_{z_i}$ an indicator for working long hours. We also include worker type ($\mathbb{1}_{x_i}$) and region ($\mathbb{1}_{s_i}$) fixed effects.

Figure 2: Returns to Hours by Occupation



Notes. This figure shows the log wage premiums for working long hours across occupations, defined as the β coefficients in equation (3.1). The data is from the [American Community Survey \(ACS, wave 2023\)](#).

Figure 2 shows the estimated long-hour premiums (β) across occupations. We find the largest premiums in Business, and particularly, for Lawyers, Judges, and Financial Operations Professionals. They earn about 6,000 to 8,000 dollars more when working long hours. The other Business occupations follow, with substantial long-hour premiums that vary between 4,000 and

5,5000 dollars. Medical Practitioners and Tech occupations have the largest wage premiums outside of Business, which vary between 3,000 and 4,500 dollars. The lowest long-hour premiums are in Administration and Training, Science, as well as the other Health occupations, where working long hours is associated with monthly earnings that are 1,000 to 2,000 dollars higher. An important thing to note is that long-hour premiums are not necessarily informative about occupational earnings. For example, Advanced Clinical Professionals have among the highest earnings, but their long hour premiums are the lowest.

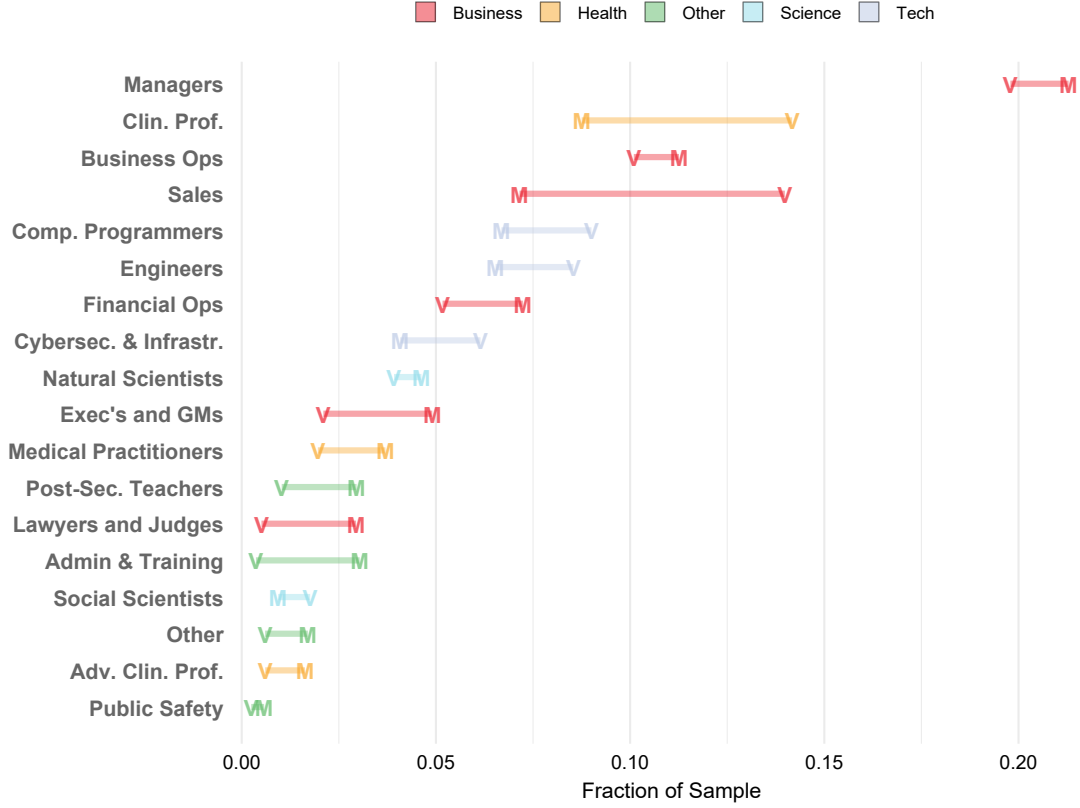
3.3 Matches and Vacancies

We now look at the distributions of matches and vacancies across occupations, as shown in Figure 3. This first notable patterns the large variation in the relative sizes of different occupations. Business occupations – particularly Managers, Sales specialists, and Business and Financial Operations – account for the largest share of both matches and vacancies, jointly making up over 45 percent of observations in each group. In contrast, Social Scientists, Public Safety Professionals, Advanced Clinical Professionals, and Other occupations together represent less than 5 percent.

A second notable pattern is the substantial difference between the distributions of vacancies and matches across occupations. In several occupations, the share of vacancies far exceeds that of matches, indicating significant excesses demand. The main examples are Clinical Professionals, Sales Specialists, and all Tech occupations, which together account for over 50 percent of vacancies but less than 33 percent of matches. This pattern aligns with [Bureau of Labor Statistics](#) (BLS) projections of occupational growth ([Bureau of Labor Statistics, 2024](#)).⁴ Conversely, several occupations show the opposite imbalance, with substantially more matches than vacancies. This includes both high-earning occupations – such as Financial Operations, Executives, Lawyers, Judges, and Medical Practitioners – and lower-earning occupations in Administration, Training, and Post-Secondary Teaching. These occupations represent roughly 25 percent of matches but only 10 percent of vacancies. Together, these patterns highlight the importance of separately estimating the occupational vacancy distribution when analyzing the market equilibrium.

⁴They include Nurses, Data scientists, and Information Security Analysts among the 5 fastest growing occupations.

Figure 3: Matches and Vacancies by Occupation



Notes. This figure highlights the occupational distributions of matches (M) and vacancies (V). Constructed using the [National Labor Exchange](#) (NLx) Research Hub and [American Community Survey](#) (ACS) data.

3.4 Wages, Hours, and the Tightness

Finally, we examine how hours and wages relate to labor market tightness, which we measure as the occupation- and worker-type specific vacancy-to-unemployment ratio. Specifically, we estimate regressions of the form:

$$\log(W_i) = \alpha + \tau_w \log(t_i) + \sum_x \mathbb{1}_{x_i} \gamma_x + \sum_y \mathbb{1}_{y_i} \gamma_y + \sum_s \mathbb{1}_{s_i} \gamma_s + \varepsilon_i, \quad (3.2)$$

$$\log(H_i) = \alpha + \tau_h \log(t_i) + \sum_x \mathbb{1}_{x_i} \gamma_x + \sum_y \mathbb{1}_{y_i} \gamma_y + \sum_s \mathbb{1}_{s_i} \gamma_s + \varepsilon_i, \quad (3.3)$$

where W_i are wages, H_i are weekly hours, and t_i is the ratio of type y vacancies to unemployed workers of type x within region s .⁵ Both regressions include fixed effects for worker types ($\mathbb{1}_{x_i}$),

⁵Similar results obtain when we replace weekly hours with an indicator for working long hours.

occupation types ($\mathbb{1}_{y_i}$), and regions ($\mathbb{1}_{s_i}$). The parameters of interest (τ_w, τ_h) are thus identified from within-type variation in tightness across regions.

Note that we observe neither the search behavior of the unemployed nor the targeting decisions of employers when they post vacancies. Workers or firms may sort into specific sub-markets based on unobserved preferences, or higher wage markets may simply attract more applicants. To address these endogeneity problems, we construct a leave-one-out instrument t_{-i} to remove the influence of local market conditions and isolate plausibly exogenous variation in supply and demand at the national level.

Table 2: Wages, Hours, and Tightness

	$\log(W_i)$	$\log(H_i)$	$\log(W_i)$	$\log(H_i)$
	OLS		IV	
$\log(t_i)$	0.017 (0.005)	-0.018 (0.005)	0.023 (0.007)	-0.031 (0.007)
Constant	8.771 (0.008)	1.849 (0.008)	8.774 (0.009)	1.842 (0.009)
Occupation FE	✓	✓	✓	✓
Worker Type FE	✓	✓	✓	✓
Region FE	✓	✓	✓	✓
First Stage F			282, 208	282, 208
R ²	0.180	0.086	0.180	0.086
Num. obs.	274, 544	274, 544	274, 544	274, 544

Notes. This table reports results from estimating equations (3.2) and (3.3) with ordinary least squares. The data is from the [American Community Survey](#) (ACS) and [National Labor Exchange](#) (NLx) Research Hub. Bold faced estimates are significant at the 5% level.

Table 2 presents the results. The first column shows an elasticity of wages with respect to tightness of 0.017, which increases to 0.023 when using the instrumental variable, as seen in the third column. A positive sign is consistent with workers having more bargaining power in tighter markets. The magnitude and upward adjustment upon instrumenting is also comparable to recent findings by [Hoeck \(2022\)](#) and [Börschlein et al. \(2024\)](#). They use similar identification strategies to estimate the elasticity in Europe, and find values between 0.01 and 0.02. The effect is also consistent with the results of [Hagedorn and Manovskii \(2008\)](#) for the United States, who identify the elasticity through the co-movement of wages and market tightness over time.

The second column of Table 2 shows an opposite elasticity of hours with respect to tight-

ness of comparable magnitude. The baseline elasticity in the second column is -0.018, which increases to -0.031 when instrumenting tightness, as shown in the fourth column. A negative sign is again consistent with workers having more bargaining power in tighter markets, allowing them to negotiate contracts with fewer hours. While there is no direct comparison for these estimates, recent work by [Autor et al. \(2023\)](#) and [Barrero et al. \(2023\)](#) highlights that labor market outcomes beyond wages respond to changes in tightness.

Table 3: Long-Hour Premiums and Tightness

	$\log(W_i)$	$\log(W_i)$
	OLS	IV
$\log(t_i)$	0.022 (0.006)	0.010 (0.008)
$\log(H_i)$	0.184 (0.002)	0.187 (0.002)
$\log(t_i) \times \log(H_i)$	-0.001 (0.002)	0.009 (0.002)
Constant	8.431 (0.009)	8.429 (0.009)
Occupation FE	✓	✓
Worker Type FE	✓	✓
Region FE	✓	✓
R ²	0.207	0.207
Num. obs.	274,544	274,544

Notes. This table reports results from estimating equation (3.2) using ordinary least squares, with an additional hours control and interaction with tightness, measured as the number of vacancies per unemployed worker by type. The data is from the [American Community Survey \(ACS\)](#) and [National Labor Exchange \(NLx\)](#) Research Hub. Bold faced estimates are significant at the 5% level.

To study the relation between long-hour premiums and tightness, we extend equation (3.2) by including hours and an interaction between hours and the vacancy-to-unemployment ratio. Table 3 shows the results. We are mainly interested in the interaction, which captures how changes in tightness affect the long-hour premium. The baseline regression in the first column has an interaction close to zero that is statistically insignificant. But the effect again increases considerably when we instrument tightness. The second column shows a significant and positive interaction that is close to 0.01, which is almost identical to the direct effect. This suggests

that a large part of the wage response to an increase in tightness occurs in markets for longer hour jobs. One interpretation of the downward bias in the baseline results is that workers with stronger preferences for working long hours select into tight high-wage markets. This motivates a search and matching framework that explicitly models these choices.

4 Directed Search and Matching with Endogenous Hours

4.1 Utility and Profit Functions

To better understand the relation between hours, wages, and market tightness, we introduce a tractable equilibrium search and matching framework with two-sided heterogeneity. Consider a labor market populated by workers $i \in I$ and firms $j \in J$, who can match one-to-one or remain unassigned.⁶ Workers are grouped into types $x = (1, 2, \dots, X)$, firms into types $y = (1, 2, \dots, Y)$, and hours arrangements into types $z = (1, 2, \dots, Z)$. All types are discrete and can be multidimensional.

Labor Supply. We model the utility of worker i of type x searching for a type y job with z hours as the sum of a systematic component and an idiosyncratic search cost:

$$u_{x_i y z} = p(\theta_{xyz})(\alpha_{xyz} + W_{xyz}) + \xi_{x_i y z}. \quad (4.1)$$

The systematic term is the product of a worker's job finding probabilities $p(\theta_{xyz})$ and their valuation of the job, which is the sum of amenities α_{xyz} and wages W_{xyz} . Job finding probabilities are a Constant Elasticity of Substitution (CES) function of market tightness as in [Menzio and Shi \(2010\)](#):⁷

$$p(\theta_{xyz}) = (1 + \theta_{xyz}^\lambda)^{\frac{1}{\lambda}}, \quad (4.2)$$

⁶In one-to-one matching models, job and firm are interchangeable. This is equivalent to firms hiring multiple workers if the output is separable across jobs.

⁷We also make a standard constant returns to scale assumption on the matching function ([Chade et al., 2017](#)).

where market tightness θ_{xyz} is the ratio of total firms to workers searching in the market, and λ is the elasticity of substitution. We assume that λ is negative, which assures that the job finding probability function maps to the unit interval and is concave increasing in tightness. The search costs ξ_{xyz} reflect idiosyncratic differences in application time, effort, or connections across different types of jobs. They rationalize why identical workers sort into different job types or hours arrangements.

Labor Demand. The profit functions of firms consist of a systematic component and an idiosyncratic screening cost:

$$\pi_{xyz} = q(\theta_{xyz})(\rho_{xyz} - W_{xyz}) + \eta_{xyz}. \quad (4.3)$$

The systematic term is the product of firms' hiring probabilities and workers' productivity net of wages. Hiring probabilities are a decreasing function of market tightness, and are defined by $q(\theta_{xyz}) = \frac{p(\theta_{xyz})}{\theta_{xyz}}$. When hired, workers generates a value ρ_{xyz} for the firm and are paid a wage W_{xyz} . The screening costs η_{xyz} capture idiosyncratic differences in compatibility or recruitment networks.

4.2 Equilibrium Wages and Flexibility

The equilibrium matches and wages are determined as the outcome of an implicit two stage game. In the first stage, each firm j posts a vacancy. They are willing to hire different types of applicants in different hours arrangements, and post wages accordingly. Job offers are posted with full commitment, and firms hold correct beliefs about the number of applying workers. The vacancies they post solve a trade-off between higher profits and the probability of hiring a worker. In the second stage, workers arrive with complete information about the available offers. They apply to one position based on an analogous trade-off between a more preferred job and the likelihood of getting hired. Firms hire the most profitable applicant. Because all workers apply simultaneously, there are coordination failures, which explain why vacant jobs and unemployment coexist in equilibrium.

Wages. We solve for wages by writing the equilibrium problem as firms maximizing profits (4.3) subject to offering workers their expected equilibrium utilities (4.1)⁸:

$$\begin{aligned} \max_{W_{xyz}, \theta_{xyz}} \quad & q(\theta_{xyz})(\rho_{xyz} - W_{xyz}) + \eta_{xyz} \\ \text{s.t.} \quad & u_{xyz} = p(\theta_{xyz})(\alpha_{xyz} + W_{xyz}). \end{aligned} \tag{4.4}$$

We substitute in the constraint and solve for equilibrium market tightness θ_{xyz}^* , see Appendix A.1. This allows us to recover a familiar equilibrium wage function W_{xyz}^* that splits up the total match surplus $\Phi_{xyz} := \alpha_{xyz} + \rho_{xyz}$. Simplifying notation, the wage function writes:

$$W_{xyz} = [1 - \mathcal{F}(\theta_{xyz})] \rho_{xyz} - \mathcal{F}(\theta_{xyz}) \alpha_{xyz}, \tag{4.5}$$

with $\mathcal{F}(\theta_{xyz})$ the elasticity of the job finding probability function with respect to market tightness:

$$\mathcal{F}(\theta_{xyz}) := \theta_{xyz} \frac{p'(\theta_{xyz})}{p(\theta_{xyz})}. \tag{4.6}$$

Equilibrium wages (4.5) increase in productivity ρ_{xyz} due to competition between firms, and decrease with better amenities α_{xyz} due to competition between workers. The extent to which these forces matter is governed by tightness in the labor market, through the elasticity of the matching function $\mathcal{F}(\theta_{xyz}) \in (0, 1)$. When there is slack in the market, $\mathcal{F}(\theta_{xyz})$ is large, and firms capture a large part of the surplus due to their superior bargaining position. In the limit, they will post wages at exactly $-\alpha_{xyz}$, only just rewarding workers for their disutility of employment. When markets become tighter, $\mathcal{F}(\theta_{xyz})$ decreases, as many firms compete for a limited number of workers. This pushes up wages, and in the limit allows workers to capture the entire value of production ρ_{xyz} .⁹

Long-Hour Premium. Let z denote a long hour arrangement and z' a full time alternative. The long-hour wage premium across occupations is defined by the expected wage difference $\Delta W_y =$

⁸Firms also maximize w.r.t. worker types x and hours z , which we will solve for in the next step.

⁹We assume throughout that $\rho_{xyz} > -\alpha_{xyz}$ such that there are systematic gains from matching.

$\mathbb{E}_x [W_{xyz} - W_{xyz'}]$, which can be decomposed into:

$$\Delta W_y = \underbrace{\mathbb{E}_x [\rho_{yz} - \rho_{yz'}]}_{\Delta \text{ Productivity}} - \underbrace{\mathbb{E}_x [F(\theta_{yz})\Phi_{yz} - F(\theta_{yz'})\Phi_{yz'}]}_{\Delta \text{ Bargained Surplus}}. \quad (4.7)$$

The equilibrium long-hour premium reflects the contributions of two components. The first captures differences in firms' production technologies, which is the channel emphasized in [Goldin \(2014\)](#). Wage premiums increase the more productive long hours are compared to full-time arrangements ($\rho_{yz} > \rho_{yz'}$). The second captures differences in the bargained surplus. Wage premiums decrease when firms capture a larger share of the long-hour surplus relative to the fraction they capture of the full-time surplus ($F(\theta_{yz})\Phi_{yz} > F(\theta_{yz'})\Phi_{yz'}$).

Matching Patterns. We can substitute equilibrium wages into the primitives of the model to obtain the following indirect utility and profit functions:

$$\bar{U}_{xyz} = p(\theta_{xyz})\mathcal{E}_{xyz}\Phi_{xyz} + \xi_{xyz}, \quad (4.8)$$

$$\bar{\Pi}_{xyz} = q(\theta_{xyz})\mathcal{F}_{xyz}\Phi_{xyz} + \eta_{xyz}, \quad (4.9)$$

where $\mathcal{E}_{xyz} = 1 - \mathcal{F}_{xyz}$. Utility and profits increase in the fraction of the joint surplus captured by workers ($\mathcal{E}_{xyz}\Phi_{xyz}$) and firms ($\mathcal{F}_{xyz}\Phi_{xyz}$) respectively.

Equations (4.8) and (4.9) define a simultaneous two sided discrete choice model, similar to that in [Choo and Siow \(2006\)](#). We follow them in assuming that shocks on both sides are iid Type 1 Extreme Value distributed.¹⁰ Utility and profit maximization with respect to partner types and

¹⁰Recent work by [Galichon and Salanié \(2022\)](#) shows that this assumption can be relaxed. [Salanié \(2015\)](#) and [Fox et al. \(2018\)](#) discuss identification of the distribution of unobserved heterogeneity, by exploiting the distribution of transfers within types or using variation across independent markets.

hours then implies closed form labor supply ($S_{yz|x}$) and demand ($D_{xz|y}$) functions:

$$S_{yz|x} := \exp(p(\theta_{xyz})\mathcal{E}_{xyz}\Phi_{xyz}) a_x, \quad (4.10)$$

$$\text{where: } a_x = \left(\sum_{y,z} \exp(p(\theta_{xyz})\mathcal{E}_{xyz}\Phi_{xyz}) \right)^{-1},$$

$$\text{and } D_{xz|y} := \exp(q(\theta_{xyz})\mathcal{F}_{xyz}\Phi_{xyz}) b_y, \quad (4.11)$$

$$\text{where: } b_y = \left(\sum_{x,z} \exp(q(\theta_{xyz})\mathcal{F}_{xyz}\Phi_{xyz}) \right)^{-1}.$$

These functions allow us to recover the matching patterns predicted by the model. Let M_{xyz} denote the predicted number of (x, y, z) matches, M_{x0} the number of type x unemployed workers, and M_{0y} the number of type y vacant positions. The equilibrium matching patterns in the model are defined by:

$$M_{xyz} = p(\theta_{xyz})S_{xyz} = q(\theta_{xyz})D_{xyz}, \quad (4.12)$$

$$M_{x0} = (1 - p(\theta_{xyz})) S_{xyz}, \quad (4.13)$$

$$M_{0y} = (1 - q(\theta_{xyz})) D_{xyz}. \quad (4.14)$$

Equilibrium Hours. Equation (4.12) is useful to study how equilibrium hours are determined across occupations. We compare again two arrangements z and z' in occupation y , and consider the worker's perspective by starting from the labor supply function.¹¹ The log relative number of long hour contracts z is defined by $\Delta M_y = \mathbb{E}_x \left[\log \left(\frac{M_{xyz}}{M_{xyz'}} \right) \right]$, which can be decomposed into:

$$\Delta M_y = \mathbb{E}_x \underbrace{[\ln p(\theta_{yz}) - \ln p(\theta_{yz'})]}_{\Delta \text{ Job Finding}} + \mathbb{E}_x \underbrace{[p(\theta_{yz})\mathcal{E}_{yz}\Phi_{yz} - p(\theta_{yz'})\mathcal{E}(\theta_{yz'})\Phi_{yz'}]}_{\Delta \text{ Expected Value Worker}}. \quad (4.15)$$

There are two forces that shape equilibrium hours in the labor market. The first are differences in the job finding rates, reflecting that there will be more matches at a given number of hours when the job finding probabilities are higher ($p(\theta_{yz}) > p(\theta_{yz'})$). The second term captures differences in the expected surplus. Workers sort into their most suitable arrangement, and if their expected

¹¹An analogous decomposition from the firm's perspective can be derived from $M_{xyh} = q(\theta_{xyz})D_{xyz}$.

surplus is larger when working longer hours ($p(\theta_{yz})\mathcal{E}_{yz}\Phi_{yz} > p(\theta_{yz'})\mathcal{E}(\theta_{yz'})\Phi_{yz'}$) this will reflect in more matches in equilibrium.

Characterizing Equilibrium. The following adding-up restrictions hold in equilibrium:

$$\begin{aligned} \sum_y \sum_z D_{xyz} &= n_x \quad \text{for all } x \in X \\ \sum_x \sum_z S_{xyz} &= f_y \quad \text{for all } y \in Y, \end{aligned} \tag{4.16}$$

where n_x denotes the total number of type x workers and f_y the total type y firms. Together with an equation for market tightness derived from the supply and demand functions, the system obtained by substituting the supply and demand functions into this set of adding-up constraints exhausts the equilibrium conditions of the model:

$$\sum_y \sum_z \exp(p(\theta_{xyz})\mathcal{E}_{xyz}\Phi_{xyz}) a_x = 1 \quad \text{for all } x \in X, \tag{4.17}$$

$$\sum_x \sum_z \exp(q(\theta_{xyz})\mathcal{F}_{xyz}\Phi_{xyz}) b_y = 1 \quad \text{for all } y \in Y, \tag{4.18}$$

$$\text{where: } \theta_{xyz} = \exp([q(\theta_{xyz})\mathcal{F}_{xyz} - p(\theta_{xyz})\mathcal{E}_{xyz}]\Phi_{xyz}) \frac{f_y b_y}{n_x a_x}. \tag{4.19}$$

We prove existence of a unique solution to this system in Appendix A.2. To do so, we first show that there is a unique solution to the market tightness equation (4.19). This solution can then be plugged into the mappings (4.17) and (4.18). We then show that this is a contracting system of $X + Y$ equations in the same number of unknowns a_x and b_y . This also highlights the approach we take to computing the equilibrium solution. This computation algorithm is iterative, starting from an arbitrary guess on market tightness, for which we solve the system defined by (4.17) and (4.18) for a_x and b_y . These new values allow us to update the guess for market tightness through (4.19), for which we can again solve the system of equations. These steps are repeated until a convergence criterion is reached.

5 Identification and Estimation

We use individual level cross-sectional data from several independent and perfectly segregated labor markets $s \in S$. In each market, we compute matching patterns by type, denoted $\hat{M}_{xyz}^s$ for observed (x, y, z) matches, $\hat{M}_{x0}^s$ for unemployed workers of type x , and $\hat{M}_{0y}^s$ for vacancies of type y . We also observe a noisy measure of wages $\hat{W}_{ijk}^s$, which we assume relates to the true equilibrium wage by:

$$\hat{W}_{ijk}^s = W_{xyh}^s + \delta_{ijk}^s \quad \text{with} \quad \delta_{ijk}^s \sim \mathcal{N}(0, \sigma) \quad \text{iid}, \quad (5.1)$$

where δ_{ijk}^s is a centered Gaussian measurement error with standard deviation σ .

Identification. The set of parameters in the model are workers' amenity valuations α_{xyz} , firms' productivity values ρ_{xyz} , the matching function elasticity λ , and the standard deviation of the wage error σ . Conditional on the elasticity, identification is similar to the frictionless model of [Dupuy and Galichon \(2022\)](#). Both labor supply and demand increase in the joint surplus Φ_{xyz} , so equilibrium matching patterns inform the model about the joint match surplus. Because we also observe transfers, we can separate out the amenity from the productivity value. We can see this in the equilibrium wage function (4.5), which increases in productivity ρ_{xyz} but decrease in amenities α_{xyz} .

With just a single market, the productivity and amenity parameters exhaust all data moments on wages and matching patterns. Similar to recent work by [Chiappori et al. \(2017\)](#) and [Fox et al. \(2018\)](#), we leverage variation across markets to identify additional parameters.¹² We assume that all markets share the same preference structure, and exploit variation in the distributions of worker- and firm types. These compositional differences lead to distinct equilibrium matching patterns and wages. Even with identical preferences, exogenous variation in the marginal distributions of workers and firms shift market tightness, and through the nonlinear matching function they generate observable differences in outcomes.¹³ Since there is only a single pa-

¹²See also the discussion in Appendix C of [Galichon et al. \(2019\)](#) on identifying the Pareto frontier curvature in matching problems with imperfectly transferable utility. As discussed before, our model also belongs to this class, and the elasticity parameter shapes the curvature.

¹³Several approaches to identify additional parameters in similar models have been proposed. [Graham et al. \(2014\)](#) introduce information about the joint surplus. [Dupuy et al. \(2023\)](#) use measures of firm-side productivity. [Salanié](#)

parameter in the matching function, the model is identified with just two markets. We prove the identification in Appendix A.3.

Separability. The transparent identifying equations follow from our assumption on the nature of the shocks. We assume that $\xi_{x_i y z}$ (and $\eta_{x y_j z}$) are independent of the *particular* firm of a given type that the worker matches with, and vice versa. This is the separability restriction introduced in Choo and Siow (2006). It is a strong assumption, because it rules out interactions between the unobservables on both sides of the market in joint surplus creation. But note that we place no restrictions on the match surplus Φ_{xyz} . The model thus performs better when detailed information on worker and firm types is available.

Parameterization. With data from multiple markets, the model is identified non-parametrically. We nonetheless introduce a flexible parametric specification for the amenity and productivity functions, reducing the computational burden and facilitating interpretation. The value of amenities is parameterized as:

$$\alpha_{xyz} = \sum_{\bar{x}, y} \mathbb{1}_{\alpha_{\bar{x} \times y}} + \sum_{x, z} \mathbb{1}_{\alpha_{x \times z}}. \quad (5.2)$$

We allow for men and women ($\bar{x}$) to have different preferences over occupations through the indicators $\mathbb{1}_{\alpha_{\bar{x} \times y}}$. These may be driven by unobserved differences in amenities or tasks over which their preferences diverge. We also allow for a type-specific valuation of the different hours options, through indicators $\mathbb{1}_{\alpha_{x \times z}}$.

The occupational production functions are parameterized as:

$$\rho_{xyz} = \sum_{\tilde{x}, y} \mathbb{1}_{\rho_{\tilde{x} \times y}} + \sum_{y, z} \mathbb{1}_{\rho_{y \times z}}. \quad (5.3)$$

We allow workers of different ages and education levels ($\tilde{x}$) to differ in their productive value across occupations, as captured by the indicators $\mathbb{1}_{\rho_{\tilde{x} \times y}}$. The productive value of different hours options is also allowed to be occupation-specific, as captured by the indicators $\mathbb{1}_{\rho_{y \times z}}$.

(2015) shows how to leverage within-type variation in wages. Ciscato (2024) discusses identification from panel data.

Estimation. We estimate the parameters of the model, denoted Ω below, using maximum likelihood. The likelihood function consist of three components. First, the contributions of employed individuals equate predicted matching patterns and wages to their sample counterparts. Second, the contributions of unemployed individuals equate predicted and observed unemployment patterns. Finally, the contributions of vacant positions equate predicted and observed vacancies. Note that we need to integrate out over the unobserved search decisions of unmatched workers and firms. The log likelihood of an individual observation can thus be expressed as:

$$\log \ell_{ijk}^s(\Omega) = \underbrace{\mathbb{1}_{xyz}^s \left[\log \tilde{M}_{xyz|\sigma}^s - \frac{(\hat{W}_{ijk}^s - W_{xyz|\Omega}^s)^2}{2\sigma^2} - \log \sigma \right]}_{\text{Matched}} + \underbrace{\mathbb{1}_{x0}^s \log \tilde{M}_{x0|\Omega}^s}_{\text{Unemployed}} + \underbrace{\mathbb{1}_{0y}^s \log \tilde{M}_{0y|\Omega}^s}_{\text{Vacant}}, \quad (5.4)$$

where $\mathbb{1}_{xyz}^s$ is an indicator that equals one if observation (i, j, k) is a match of type (x, y, z) in market s , $\mathbb{1}_{x0}^s$ is an indicator for observation i being an unemployed person of type x in market s , and $\mathbb{1}_{0y}^s$ is an indicator for observation j being a vacant position of type y in market s . The estimated parameter vector $\hat{\Omega}$ solves:

$$\hat{\Omega} = \arg \max_{\Omega} \left\{ \sum_{s,i,j,k} \log \ell_{ijk}^s(\Omega) \right\}. \quad (5.5)$$

Scaling. The estimation relies on multiple cross-sections of matches, unemployment, and vacancy stocks. But these stocks differ in magnitude from the typical implied steady state hiring and separation flows. To reconcile the large number of matches with low unemployment and vacancy rates, the model would drive up the matching function elasticity λ . This leads to unstable predictions, where small surplus differences have large effects on matches and wages. We avoid this by re-scaling the unemployment stocks, reducing the implied job-finding rate p^* to 27.5%.¹⁴ The stock of vacancies is scaled to match the aggregate job openings to unemployment ratio from the [Job Openings and Labor Turnover Survey \(JOLTS\)](#). This does not alter the likelihood function, it just changes the inputs. Further details are provided in [Appendix A.4](#).

¹⁴We choose 27.5% to reflect recent estimates. [Eeckhout and Lindenlaub \(2019\)](#) and [Eeckhout and Sepahsalari \(2024\)](#) use 25%, and [Barnichon and Figura \(2015\)](#) estimate that the job finding rate of highly educated workers is roughly 10% higher. The exact rate is inconsequential, and simply assures the model does not drive up the matching function elasticity too much.

6 Results

This section first presents the main parameter estimates for firms' production technologies and workers' preferences over hours. We then study the distribution of bargaining weights across occupations. After that, we compare the model's wage premium to that in the data, and decompose it into a difference in productivity and bargained surplus. Finally, we study how reducing the productivity premium to long hours affects gender equality in compensation.

6.1 Valuations of Hours

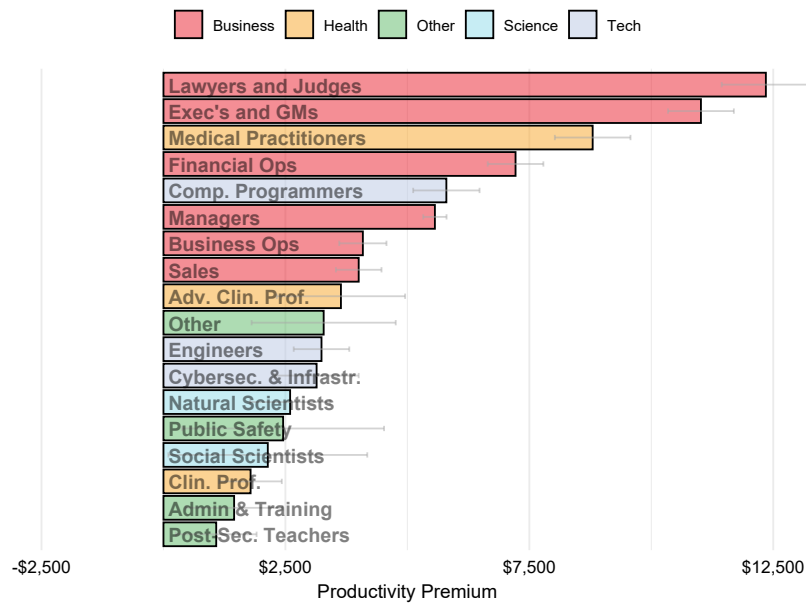
The Productivity of Hours. Panel (a) of Figure 4 highlights the productivity premium to working long hours across occupations. The first main result is that the large long-hour wage premiums that we documented for Business occupations reflect substantial differences in their production functions. This confirms the assertion of Goldin (2014), who built her narrative based only on observed equilibrium wage patterns. We find the largest premiums for Lawyers, Judges, and Executives, who generate an additional 10,000 to 12,500 dollars per month when working long hours as opposed to full-time. The other Business occupations follow, such as Financial Operations and Managers, generating an additional 5,000 to 7,500 dollars per month, and Managers, Business Operations, and Sales, generating an additional 3,500 and 6,000 dollars.

At the bottom end of the productivity premium distribution, we generally find the occupations for which we also document lower wage premiums. These are occupations in Science, Teaching, Training or Clinical Professionals. Their productivity premiums are all between 1,000 and 2,500 dollars. Two important exceptions are Advanced Clinical Professionals and Medical Practitioners, who generate substantial value by working longer hours, at almost 3,500 and 9,000 dollars per month respectively. But this is not reflected in their wage structures. To a lesser extent, this also holds for Computer Programmers. Understanding what shapes their wage profiles requires us to look beyond immediate productive differences.

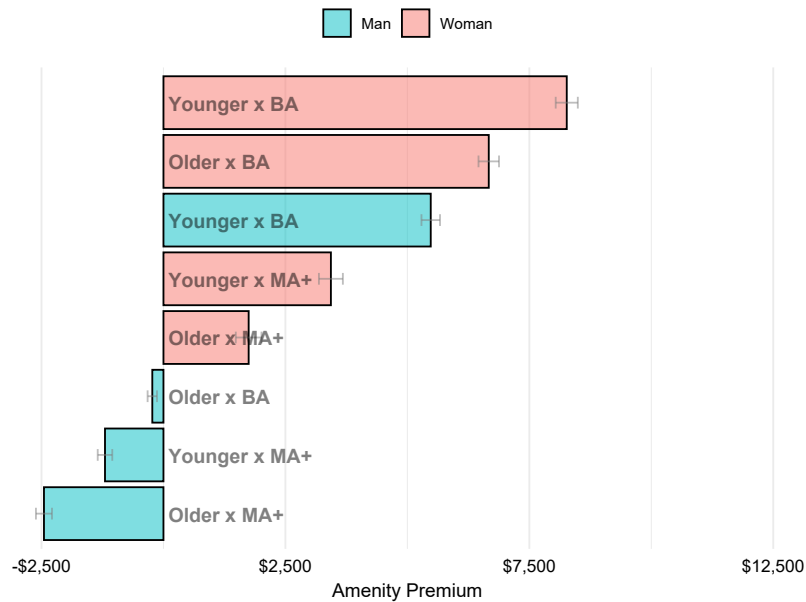
Preferences for Hours. Panel (b) highlights preferences for hours across different types of workers. We find that women on average always prefer working full-time over long hours. This is particularly true for those without a Master's degree, who on average require wage premiums

Figure 4: The Value of Hours

(a) Productivity Premium by Occupation



(b) Amenity Premium by Worker Type



Notes. This figure shows the estimated productivity premiums for working long hours across occupations in Panel (a) and the amenity value of hours across worker types in Panel (b). The value created by working full-time is normalized to zero on the firm side, so these parameters reflect the additional value created by working long hours across occupations. The amenity value of long-hours is are normalized to zero on the worker side. All values are expressed in dollars of monthly earnings, and the grey lines indicate 95% confidence bounds.

between 6,000 and 8,000 dollars. On the other hand, men generally prefer working longer hours. Particularly, those with at least a Master’s degree would forego between 1,000 and 2,500 dollars per month to work a long-hour job. Given their high earnings, this represents an 8 to 15% reduction in average monthly earnings. Their preferences are likely driven by both dynamic incentives and prestige. A recent paper by [Jang and Yum \(2022\)](#) highlights the importance of the former when studying long hours of work.

Further Estimates. The other parameters can be found in Appendix [C](#). One important parameter is the elasticity of substitution λ , which we estimate to be -0.33. This is close to the values commonly used in the literature, such as [Eeckhout and Sepahsalari \(2024\)](#), who set it at -0.41. We show workers’ preferences over occupations and firms’ preferences over different types of workers. [to add]

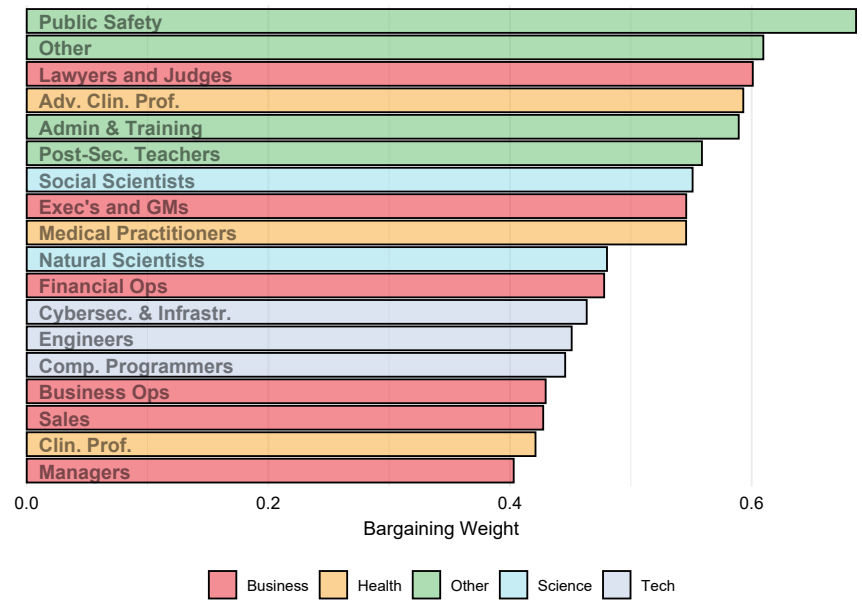
6.2 The Distribution of Bargaining Power

To better understand how tightness in the market shapes equilibrium wages, we look at the bargaining weights of firms in Figure [5](#). Panel (a) shows the median bargaining weights across occupations. The occupations with the highest number of vacancies, as shown in Figure [3](#), are characterized by firms having lower bargaining weights. We find that in the occupations of Managers, Salespeople, Business Operations, Clinical Professionals, or in Tech, the perceived bargaining weights of firms are roughly 0.40 to 0.45. When vacancies are scarce, the perceived bargaining power of firms is larger. In all of the highest earning occupations, being Lawyers and Judges, Executives, and Medical Practitioners, firms have bargaining weights between 0.55 and 60. The weights are similar for Post-Secondary Teachers, Administration and Training, Advanced Clinical Professionals, Social Science, and Public Safety occupations, which are all characterized by having little vacancies.

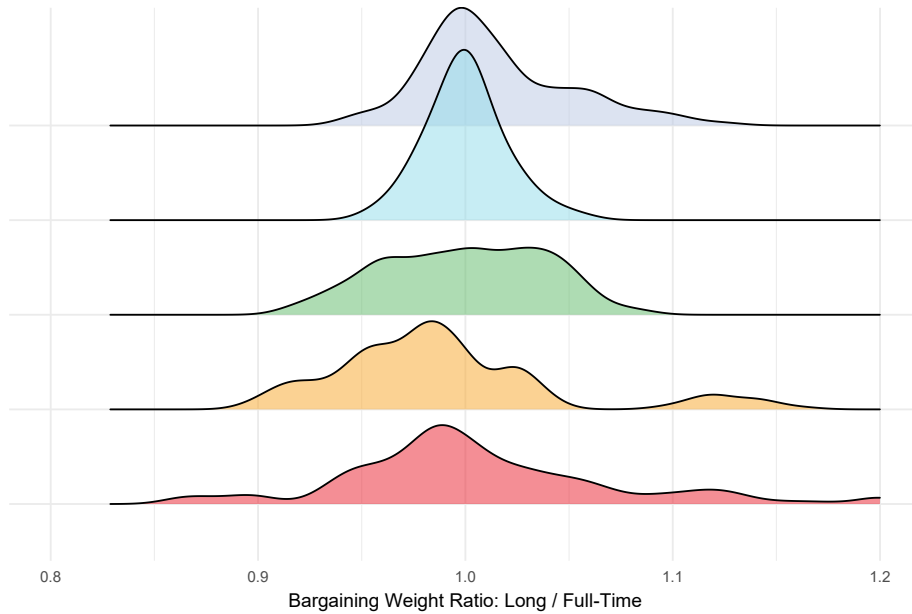
To better understand how these bargaining weights are distributed, and differ between the long-hour and full-time market, we compute for each occupation the ratio’s $\left(\mathcal{F}_{xyz}^s / \mathcal{F}_{xyz'}^s\right)$. When these exceed one, firms have more bargaining power in the long-hour market compared to the corresponding full-time market, which pushes down the long-hour premium. Panel (b) highlights the distribution of these ratio’s. The first thing to note is that across all occupation groups, the

Figure 5: Bargaining Weights across Occupations

(a) Median Bargaining Weights



(b) Distributions of Bargaining Weight Ratio



Notes. This figure highlights the distribution of bargaining weights across occupations. Panel (a) shows the median bargaining weights. Panel (b) presents the distribution of the bargaining ratio, defined as firms' bargaining power in long hours over full time markets ($\mathcal{F}_{xyz}^s / \mathcal{F}_{xyz'}^s$). These distributions are winsorized at 0.8 and 1.2, with all values outside this range replaced with the respective bound.

mass of these distributions is located between 0.90 and 1.10, but the shape differs. In Tech occupations we see that firms have somewhat more bargaining power in the long-hour markets than the full-time market, which puts downward pressure on the wage premiums. We see the opposite in Health occupations, where bargaining power differences elevate the wage premiums. In Business and Other occupations the distributions are more spread, indicating relatively large effects, but there seems to be no clear upward or downward trend.

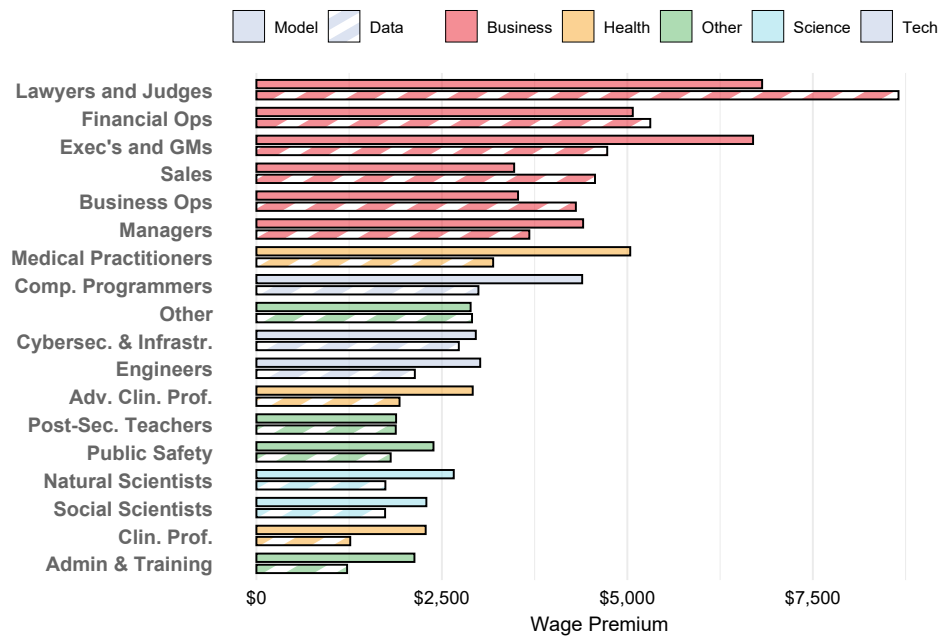
6.3 The Equilibrium Long Hour Premium.

We now look at how the model's long hour premiums fit those in data, and then decompose the premium into a productivity and bargained surplus component, as shown in equation (4.7). The model fit is highlighted in Panel (a) of Figure 6. The model generally replicates the long-hour premiums well, but the fit differs between occupations. The main disparity in Business occupations is that the model underestimates the long-hour premiums of Lawyers and Judges, and overestimates the premium for Executives and GMs. The model also overestimates the long-hour premium in Health and Tech occupations, notably for Computer Programmers and Medical Practitioners. There are some differences between the data and model for other occupations, but overall they model replicates the main patterns in the data.

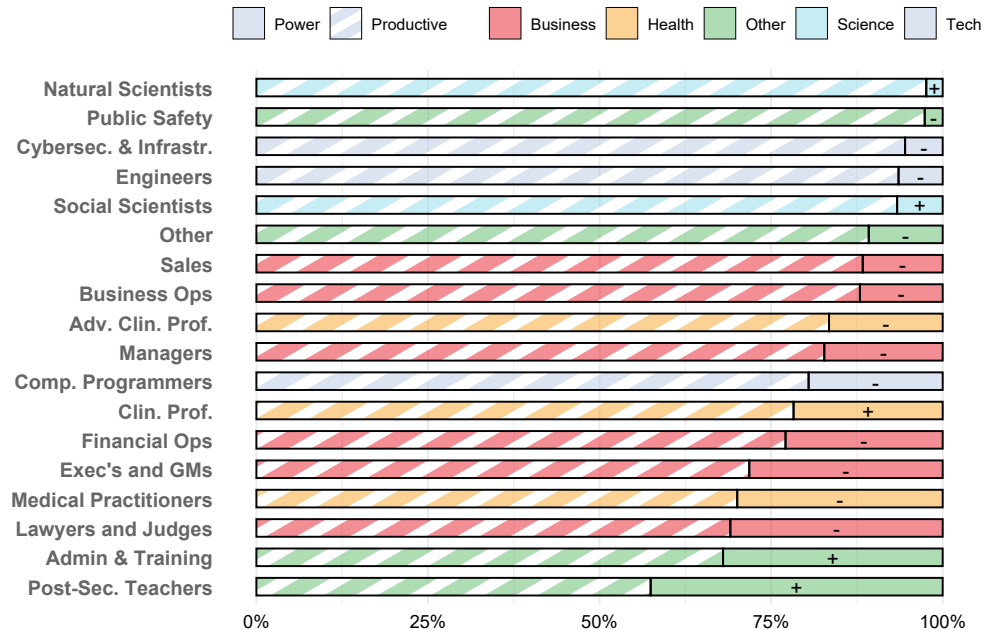
Decomposition. As a next step, we decompose the long-hour premiums into productivity and bargained surplus differences. Panel (b) of Figure 6 shows the percentage contribution by each term. The signs within each bar indicate whether the bargaining difference inflates or reduces the long-hour premium. The first thing to note is that productive differences explain the majority of the premium. This is again in line with the narrative of Goldin (2014). On average, we estimate that productive differences contribute 82%, while 18% is due to bargaining different fractions of the surplus. But the average masks significant heterogeneity across occupations, both in the magnitude and direction of the effect. In Administration, Post-Secondary Teaching, and Clinical Professional occupations, workers are able to capture a larger part of the surplus in the long-hour market than in the full-time market. This inflates wage premiums by up to 40%. At the other end, wage premiums in the highest earning occupations in our sample are pushed down, as workers capture a significantly smaller fraction of the long-hour surplus. The long hour premium

Figure 6: Wage Premium Decomposition

(a) Model vs. Data Premium



(b) Decomposition of Premium

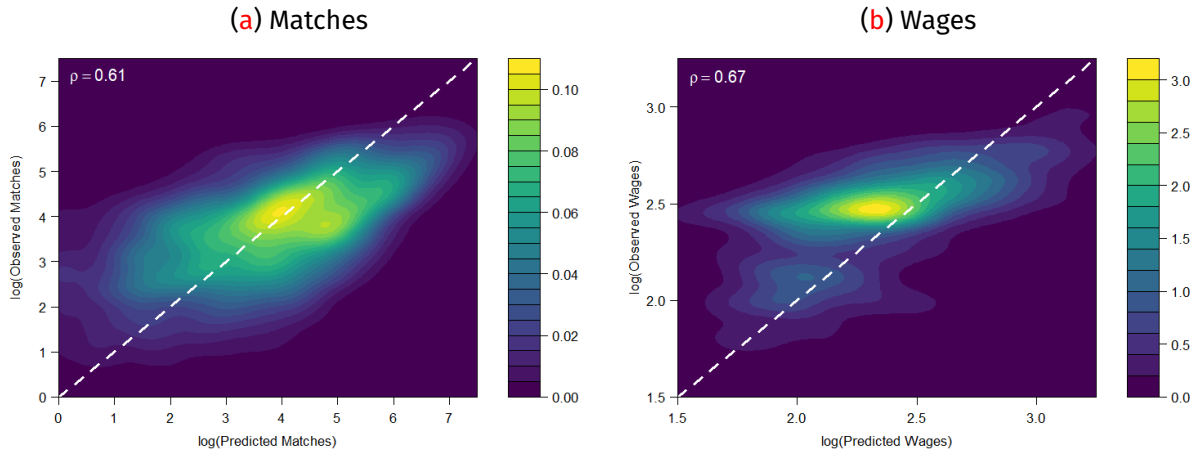


Notes. This figure highlights the fit and decomposition of the long-hour premium across occupations. Panel (a) presents the premium in the data and in the model. Panel (b) shows the decomposition into productivity and tightness related components. The signs indicate whether the power component increases (+) or decreases (-) the wage premium.

for Lawyers, Judges, Medical Practitioners, Executives, and GMs is reduced by about 30% due to workers capturing less of the surplus. This is in line with significant competition among workers for prestigious high earning jobs.

Fit and Individual Variation. To further assess the fit, figure 7 highlights plots the sample moments of matching patterns and wages in each sub-market ($X \times Y \times Z \times S$) against the predicted model counterparts. The diagonal line indicates a perfect model fit. Panel (a) shows that the model replicates matching patterns well. The correlation between observed and predicted matching patterns is 0.61. The main markets where the model does not fit well are those for which we have only few observations, for which we under-predict the number of matches. Panel (b) shows that the model does a better job at replicating wages, with a correlation of 0.67. In general we over-predict wages slightly, particularly at upper-intermediate levels. The model explains about 16% of individual wage variation, which is slightly less than what a wage regression explains in our sample with the same types, at 20%. But the latter only fits wages and not matching patterns.

Figure 7: Model Fit

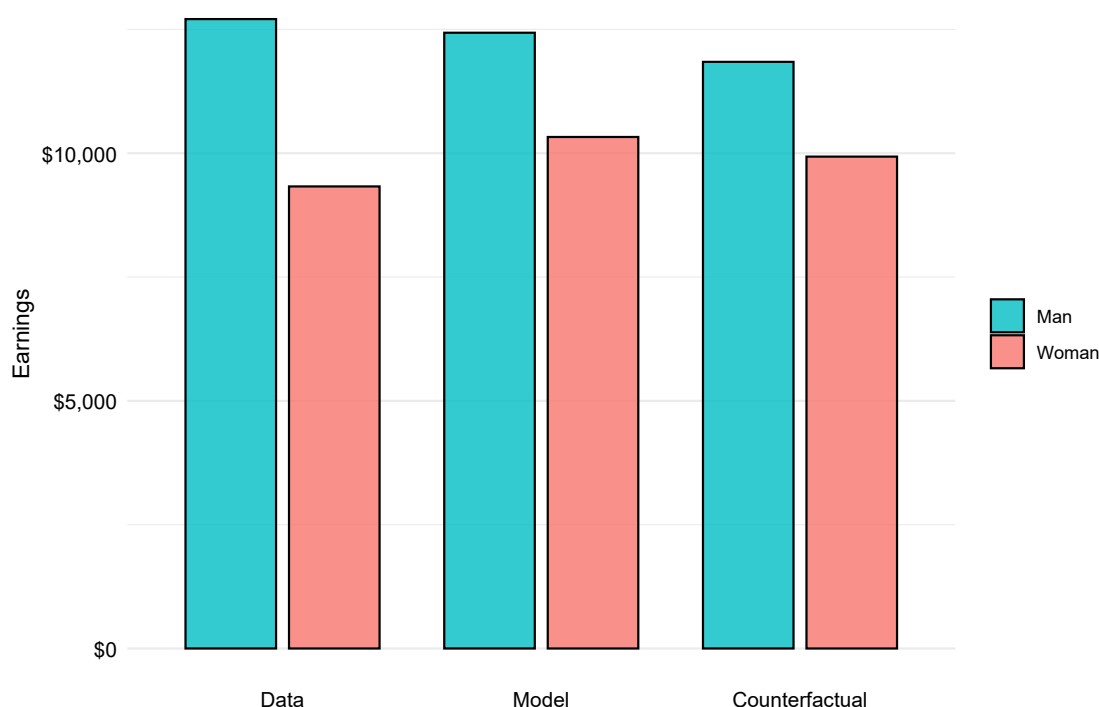


Notes. This figure plots the model predictions against the data moments. Panel (a) highlights the matching pattern fit, and Panel (b) the fit of wages. The 45° line indicates that the model exactly replicates the data. Both matching patterns and wages are expressed in logs. The color scale highlights the density of observations obtained from a kernel estimator.

6.4 The Gender Gap in Earnings: Counterfactual

The next step in the analysis is to study the implications for gender differences in earnings. [Goldin \(2014\)](#) notes that the gender gap may be considerably reduced if firms did not have incentives to disproportionately reward long and particular hours of work. Such changes have been documented for pharmacists by [Goldin and Katz \(2016\)](#), where the wage structure became linear as technological improvements made workers better substitutes. We test the hypothesis on the entire high-end labor market by halving the long-hour productivity premiums, and compute the counterfactual equilibrium matching patterns and wages.

Figure 8: Gender Gap in Earnings



Notes. This figure highlights the gender wage gap in the [American Community Survey](#) (ACS, wave 2023), the model, and a counterfactual simulation that halves the productivity premium to long hours.

Figure 8 highlights the results from this analysis. The model replicates the average wages of men and women well, but slightly over-predicts women's wages. The gender wage gap in the data is about 26%, while the model predicts a gap of about 17%. An important reason for this difference is that we do not introduce gender as a type in the firm's production function, shutting down firm-side discrimination. Recent work by [Sin et al. \(2022\)](#) argues that this can explain up

to 45% of the gender gap in wages. We then look at what happens in the counterfactual scenario where firms' incentives to reward long hours are reduced. As expected, this leads to a reduction in earnings, but this reduction larger for men. This has only a moderate effect on the gender wage gap, which reduces by about 5%.

7 Conclusion

This paper studies how hours and wages are determined in the labor market. I first empirically document that hours, wages, and equilibrium long-hour premiums all respond to changes in market tightness. To better understand how these patterns should be interpreted, I then introduce a tractable equilibrium search and matching model, which features directed search as in [Moen \(1997\)](#) while allowing for two sided multidimensional heterogeneity as in [Choo and Siow \(2006\)](#). Hours and wages are jointly determined at equilibrium, as a function of the production technologies of firms, the preferences of workers, and tightness in the labor market.

The first main result from the model is that the productivity of long working hours differs significantly across occupations. In line with the narrative of [Goldin \(2014\)](#) and [Goldin and Katz \(2016\)](#), I find that Business occupations are characterized by large productivity premiums for working long weekly hours, and that these translate into substantial equilibrium wage premiums. This is less the case for Health, Tech, Science and Other occupations. The second main result is that wage premiums are not only determined by differences in productivity. A decomposition exercise shows that about 80% of the premium arises due to differences in productivity. The other 20% stems from differences in the fraction of the surplus that firms capture when they hire a full-time as opposed to a long hour worker. There is again substantial occupational heterogeneity, in both the magnitude and sign of this effect. Finally, a counterfactual analysis shows that reducing the long-hour surplus only moderately changes the gender gap in wages, which reduces by about 5%.

To conclude, I point out some directions for future research. A first extension is to make the model dynamic, which would better capture motivations on the supply side of the labor market. This allows us to study how hours choices affect human capital accumulation across occupations over the life cycle. Another extension of the model would be to introduce households, as labor

supply choices are clearly not made in isolation. The particular benefit of this model would be that it allows us to study how the intra-household bargaining process is shaped by the structure of the labor market.

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A Model

A.1 Equilibrium Tightness and Wages

We start by writing the equilibrium problem similar to [Wright et al. \(2021\)](#) as firms maximizing profits subject to offering workers their expected equilibrium utility:

$$\begin{aligned} \max_{W_{xyz}, \theta_{xyz}} \quad & q(\theta_{xyz})(\rho_{xyz} - W_{xyz}) + \eta_{xyz} \\ \text{s.t.} \quad & u_{xyz} = p(\theta_{xyz})(\alpha_{xyz} + W_{xyz}). \end{aligned} \tag{A.1}$$

Rewriting the constraint in terms of wages and substituting it out of the objective function yields:

$$\max_{W_{xyz}, \theta_{xyz}} \left\{ \frac{p(\theta_{xyz})(\rho_{xyz} + \alpha_{xyz}) - u_{xyz}}{\theta_{xyz}} + \eta_{xyz} \right\}. \tag{A.2}$$

Taking the first order condition of equation (A.2) with respect to market tightness yields:

$$u_{xyz} = (\rho_{xyz} + \alpha_{xyz}) (p(\theta_{xyz}^*) - \theta_{xyz}^* p'(\theta_{xyz}^*)). \tag{A.3}$$

Substituting this back into the constraint yields a characterization of equilibrium wages:

$$W_{xyz}^* = (1 - \mathcal{F}(\theta_{xyz}^*)) \rho_{xyz} - \mathcal{F}(\theta_{xyz}^*) \alpha_{xyz}, \tag{A.4}$$

with $\mathcal{F}(\theta_{xyz}^*)$ the elasticity of the job finding probability function with respect to market tightness:

$$\mathcal{F}(\theta_{xyz}^*) := \theta_{xyz}^* \frac{p'(\theta_{xyz}^*)}{p(\theta_{xyz}^*)}. \tag{A.5}$$

We use expression (A.4) to substitute out wages to obtain the indirect utility (4.8) and profit (4.9) functions from section 4.2 in the main text.

A.2 Existence and Uniqueness of Equilibrium

We now establish existence of a unique equilibrium. We start from equations (4.17)–(4.19) and substitute out $p(\theta_{xyz})$, $q(\theta_{xyz})$, $\mathcal{E}_{xyz}$, and $\mathcal{F}_{xyz}$ using the matching function defined in equation (4.2). This yields the following system of $X + Y$ equations that characterizes the equilibrium:

$$\sum_{y,z} \exp \left((1 + \theta_{xyz}^\lambda)^{(1-\lambda)/\lambda} \Phi_{xyz} \right) a_x = 1 \quad \text{for all } x, \quad (\text{A.6})$$

$$\sum_{x,z} \exp \left(\theta_{xyz}^{\lambda-1} (1 + \theta_{xyz}^\lambda)^{(1-\lambda)/\lambda} \Phi_{xyz} \right) b_y = 1 \quad \text{for all } y, \quad (\text{A.7})$$

$$\text{where: } \theta_{xyz} = \exp \left(\frac{\theta_{xyz}^{\lambda-1} - 1}{(1 + \theta_{xyz}^\lambda)^{(\lambda-1)/\lambda}} \Phi_{xyz} \right) \frac{a_x f_y}{b_y n_x}. \quad (\text{A.8})$$

We want to show that there exists a unique equilibrium solution (a_x, b_y) . As in the main text we impose the restrictions that $\lambda < 0$, $\Phi_{xyz} > 0$, and both n_x and $f_y > 0$. Note also that by definition both a_x and b_y are take on values between in $(0, 1)$. We proceed by first showing that the implicit function (A.8) defining market tightness has a unique solution, and then establish uniqueness of the equilibrium system defined by (A.6) and (A.7).

Unique Root. Market tightness is strictly positive because measures the number of searching firms over searching workers, so we look for a root $(0, +\infty)$ of the function:

$$G(\theta_{xyz}) = \theta_{xyz} - \exp \left(\frac{\theta_{xyz}^{\lambda-1} - 1}{(1 + \theta_{xyz}^\lambda)^{(\lambda-1)/\lambda}} \Phi_{xyz} \right) \frac{a_x f_y}{b_y n_x}. \quad (\text{A.9})$$

At the limits, this expression yields:

$$\lim_{\theta_{xyz} \rightarrow 0} G(\theta_{xyz}) = - \exp(\Phi_{xyz}) \frac{a_x f_y}{b_y n_x} < 0,$$

$$\lim_{\theta_{xyz} \rightarrow \infty} G(\theta_{xyz}) = \theta_{xyz} - \exp(-\Phi_{xyz}) \frac{a_x f_y}{b_y n_x} = \infty.$$

Since $G(\theta_{xyz})$ is continuous, the intermediate value theorem guarantees that a root exists. To show uniqueness, we examine the derivative of $G(\theta_{xyz})$. We first rewrite the inner exponent term:

$$\psi(\theta_{xyz}) := \frac{\theta_{xyz}^{\lambda-1} - 1}{(1 + \theta_{xyz}^\lambda)^{(\lambda-1)/\lambda}} \Phi_{xyz}, \quad (\text{A.10})$$

and write the derivative of $G'(\theta_{xyz})$ as:

$$G'(\theta_{xyz}) = 1 - \exp(\psi(\theta_{xyz})) \cdot \frac{a_x}{b_y} \cdot \frac{f_y}{n_x} \cdot \psi'(\theta_{xyz}). \quad (\text{A.11})$$

Since all multiplicative terms are strictly positive, the sign of $G'(\theta_{xyz})$ depends on $\psi'(\theta_{xyz})$:

$$\psi'(\theta_{xyz}) = \Phi_{xyz} \cdot \frac{(\lambda - 1)(\theta_{xyz} + 1) \cdot \theta_{xyz}^{\lambda-2}}{(1 + \theta_{xyz}^\lambda)^{2(\lambda-1)/\lambda}} < 0. \quad (\text{A.12})$$

This term is strictly negative because $\Phi_{xyz} > 0$, $\lambda < 0$, and $\theta_{xyz} > 0$. It follows that $G'(\theta_{xyz}) > 0$, so and thus $G(\theta_{xyz})$ is strictly increasing and has a unique root. This in turn implies that there is a unique solution θ_{xyz}^* that solves (A.8) for every (x, y, z) .

Unique Equilibrium. We now consider (A.6) and (A.7) as a mapping $T : (a_x, b_y) \mapsto (a'_x, b'_y)$:

$$a'_x = \left(\sum_{y,z} \exp \left((1 + \theta_{xyz}^\lambda)^{(1-\lambda)/\lambda} \Phi_{xyz} \right) \right)^{-1} \quad \text{for all } x, \quad (\text{A.13})$$

$$b'_y = \left(\sum_{x,z} \exp \left(\theta_{xyz}^{\lambda-1} (1 + \theta_{xyz}^\lambda)^{(1-\lambda)/\lambda} \Phi_{xyz} \right) \right)^{-1} \quad \text{for all } y, \quad (\text{A.14})$$

$$\text{with: } \theta_{xyz}(a_x, b_y) = \exp \left(\frac{\theta_{xyz}^{\lambda-1} - 1}{(1 + \theta_{xyz}^\lambda)^{(\lambda-1)/\lambda}} \Phi_{xyz} \right) \frac{a_x}{b_y} \frac{f_y}{n_x}. \quad (\text{A.15})$$

The mapping T is continuous and maps strictly positive values of (a_x, b_y) into a bounded and well-behaved set of values on the $(0, 1)$ interval, since each update is the inverse of a sum of positive exponentials. Existence of a fixed point follows from Brouwer's fixed point theorem. Moreover, the exponential terms in (A.13) and (A.14) grow faster than the linear effects of a_x and b_y on θ_{xyz} . This ensures that subsequent updates of (a_x, b_y) are dampened. To show uniqueness, note first that θ_{xyz} is strictly increasing in a_x and decreasing in b_y , and update expressions are

strictly decreasing in θ_{xyz} . As a result, increasing a_x or decreasing b_y reduces a'_x and b'_y respectively. This strict monotonicity implies that the mapping T cannot intersect the identity more than once, which ensures that the fixed point is unique.

A.3 Identification

We now formally show the identification of the productivity (ρ_{xyz}) and amenity (α_{xyz}) parameters, as well as the matching function elasticity (λ). We still consider a setting with data on matching patterns and wages in multiple segmented markets $s \in S$, but omit the additional superscripts for clarity. We start by expressing the observed matching patterns and wages in terms of their nonlinear model expressions. We then show that these equations are monotonic in the structural parameters, and can thus be inverted. This establishes identification, assuming that we have data on a sufficient number of markets.

The model expressions are given by:

$$\hat{M}_{xyz}/\hat{n}_x = S_{yz|x}p(\theta_{xyz}) = \exp\left((1 + \theta_{xyz}^\lambda)^{(1-\lambda)/\lambda}\Phi_{xyz}\right) a_x(1 + \theta_{xyz}^\lambda)^{1/\lambda}, \quad (\text{A.16})$$

$$\hat{M}_{xyz}/\hat{f}_y = D_{xz|y}q(\theta_{xyz}) = \exp\left(\theta_{xyz}^{\lambda-1}(1 + \theta_{xyz}^\lambda)^{(1-\lambda)/\lambda}\Phi_{xyz}\right) b_y(1 + \theta_{xyz}^\lambda)^{1/\lambda}\theta_{xyz}^{-1}, \quad (\text{A.17})$$

$$\begin{aligned} \hat{M}_{x0}/\hat{n}_x &= S_{yz|x}(1 - p(\theta_{xyz})), \\ &= \exp\left((1 + \theta_{xyz}^\lambda)^{(1-\lambda)/\lambda}\Phi_{xyz}\right) a_x \left(1 - (1 + \theta_{xyz}^\lambda)^{1/\lambda}\right), \end{aligned} \quad (\text{A.18})$$

$$\begin{aligned} \hat{M}_{0y}/\hat{f}_y &= D_{xz|y}(1 - q(\theta_{xyz})), \\ &= \exp\left(\theta_{xyz}^{\lambda-1}(1 + \theta_{xyz}^\lambda)^{(1-\lambda)/\lambda}\Phi_{xyz}\right) b_y \left(1 - (1 + \theta_{xyz}^\lambda)^{1/\lambda}\theta_{xyz}^{-1}\right), \end{aligned} \quad (\text{A.19})$$

$$\hat{W}_{xyz} = \frac{\rho_{xyz} - \theta_{xyz}^\lambda \alpha_{xyz}}{1 + \theta_{xyz}^\lambda}. \quad (\text{A.20})$$

We still assume that $\lambda < 0$ to assure that the job- and worker finding probability functions are well-defined. We also consider the case where $\Phi_{xyz} > 0$ such that there are gains to be made from matching. We further only include types for which there are observations, so both n_x and f_y are strictly positive, as is market tightness θ_{xyh} .

Amenities and Productivity. The joint surplus Φ_{xyh} can be identified from the observed number of matches $\hat{M}_{xyz}$. In both equations (A.16) and (A.17), all terms are strictly positive on the relevant domain, and Φ_{xyz} enters linearly in the exponent. Although Φ_{xyz} also influences market tightness

θ_{xyz} , we can implicitly differentiate the tightness equation to verify that θ_{xyz} is strictly increasing in Φ_{xyz} . Recall that θ_{xyz} is implicitly defined by:

$$\theta_{xyz} = \exp [\psi(\theta_{xyz})\Phi_{xyz}] C_{xy}, \quad (\text{A.21})$$

where $\psi(\theta_{xyz})$ is a function of θ_{xyz} and C_{xy} a positive constant. Differentiating both sides with respect to Φ_{xyz} yields:

$$\frac{\partial \theta_{xyz}}{\partial \Phi_{xyz}} = \frac{\partial \exp [\psi(\theta_{xyz})\Phi_{xyz}] C_{xy}}{\partial \Phi_{xyz}} \quad (\text{A.22})$$

$$= \exp [\psi(\theta_{xyz})\Phi_{xyz}] C_{xy} \left[\psi(\theta_{xyz}) + \Phi_{xyz} \psi'(\theta_{xyz}) \frac{\partial \theta_{xyz}}{\partial \Phi_{xyz}} \right] \quad (\text{A.23})$$

We can substitute in $\theta_{xyz} = \exp [\psi(\theta_{xyz})\Phi_{xyz}] C_{xy}$ and rewrite:

$$\frac{\partial \theta_{xyz}}{\partial \Phi_{xyz}} = \theta_{xyz} \left(\psi(\theta_{xyz}) + \Phi_{xyz} \psi'(\theta_{xyz}) \frac{\partial \theta_{xyz}}{\partial \Phi_{xyz}} \right) \quad (\text{A.24})$$

$$= \frac{\psi(\theta_{xyz})\theta_{xyz}}{1 - \Phi_{xyz} \psi'(\theta_{xyz})\theta_{xyz}} > 0. \quad (\text{A.25})$$

This expression is strictly positive since $\psi(\theta_{xyz}) > 0$ and $\psi'(\theta_{xyz}) < 0$ for all $\lambda < 0$ and $\theta_{xyz} > 0$. Therefore, market tightness is strictly increasing in surplus Φ_{xyz} . Therefore, the number of matches increases monotonically in the surplus, allowing us to invert these expressions to identify the surplus. Separate identification of the amenity and productivity terms follows from the observation of wages. As shown in equation (A.20), equilibrium wages increase in productivity (ρ_{xyz}) and decrease in amenities (α_{xyz}) at a given value of tightness. Since θ_{xyz} is separately identified from the matching patterns, the wage equation provides a one-to-one mapping from $(\rho_{xyz}, \alpha_{xyz})$ to observed wages.

The Job Finding Elasticity. The parameter λ governs the curvature of the matching function, and is identified from variation across markets. We assume that the surplus terms ρ_{xyz} and α_{xyz} are invariant across markets $s \in S$, but market tightness $\theta_{xyz|s}$ varies due to differences in the marginal worker ($n_{x|s}$) and firm ($f_{y|s}$) distributions. Equation (A.20) shows that equilibrium wages

are also nonlinear function of λ . We can differentiate this expression to obtain:

$$\frac{\partial \hat{W}_{xyz|s}}{\partial \lambda} = - \frac{\theta_{xyz|s}^{\lambda} \log(\theta_{xyz|s}) (\alpha_{xyz} + \rho_{xyz})}{(1 + \theta_{xyz|s}^{\lambda})^2}. \quad (\text{A.26})$$

The sign of this derivative will depend on whether $\theta_{xyz|s} > 1$ or $\theta_{xyz|s} < 1$. But for any fixed $\theta_{xyz|s} > 0$ this function is strictly monotonic in λ . Assuming that we have multiple markets where we observe wages and matching patterns for varying marginals in a sub-market (x, y, z) , we can invert this system to recover λ . Since the model contains only a single elasticity parameter, observing $S = 2$ distinct markets suffices for identification.

A.4 Steady State Scaling

Consider a textbook discrete time [Diamond \(1982\)](#)–[Mortensen \(1982\)](#)–[Pissarides \(1985\)](#) model. Let u_t and v_t denote the unemployment and vacancy rates at time t , and let market tightness be defined as $\theta_t = \frac{v_t}{u_t}$. Matches form at flow rate:

$$M_t = p(\theta_t) u_t, \quad (\text{A.27})$$

where $p(\theta_t)$ is the job finding rate and $q(\theta_t) = \frac{p(\theta_t)}{\theta_t}$ the vacancy filling rate. Workers separate from employment at a rate δ , so unemployment evolves according to:

$$u_{t+1} = \delta(1 - u_t) + [1 - p(\theta_t)] u_t. \quad (\text{A.28})$$

In steady state $u_{t+1} = u$ and $\theta_t = \theta$, so unemployment and flow matches are given by:

$$u = \frac{\delta}{\delta + p(\theta)} \quad \text{and} \quad M = p(\theta)u. \quad (\text{A.29})$$

This implies that unemployment and vacancies are:

$$u = \frac{M}{p(\theta)} \quad \text{and} \quad v = \frac{M}{q(\theta)}. \quad (\text{A.30})$$

Scaling. In cross-sectional data we observe only the stocks of matches ($\hat{M}_{xyz}$), unemployed workers ($\hat{M}_{x0}$), and vacant jobs ($\hat{M}_{y0}$). We aggregate these into:

$$E = \sum_{x,y,z} \hat{M}_{xyz} \quad \text{and} \quad U = \sum_x \hat{M}_{x0} \quad \text{and} \quad V = \sum_y \hat{M}_{y0}. \quad (\text{A.31})$$

In steady state, the flow of matches equals the flow of separations:

$$M = \delta E. \quad (\text{A.32})$$

If we use raw stocks without adjustment, the matching efficiency is overstated by roughly $\frac{\delta+p}{p}$. This forces the estimated matching-elasticity λ to highly negative values, which leads to volatile

model predictions. To avoid this, we scale the data to match plausible steady-state job-finding and hiring rates. We choose a job finding rate of 27.5% to reflect recent estimates. [Eeckhout and Lindenlaub \(2019\)](#) and [Eeckhout and Sepahsalari \(2024\)](#) use 25%, and [Barnichon and Figura \(2015\)](#) estimate that the job finding rate of highly educated workers is roughly 10% higher. The implied vacancy filling rate $q^* = p^*/\theta$ is set to match the average vacancies per unemployment in the [Job Openings and Labor Turnover Survey \(JOLTS\)](#).

We introduce scaled stocks as:

$$U^* = k_n U \quad \text{and} \quad V^* = k_f V, \quad (\text{A.33})$$

and impose:

$$p^* = M/U^* \quad \text{and} \quad q^* = M/V^*. \quad (\text{A.34})$$

Solving for the scaling factors we get:

$$k_n = \frac{\delta E}{p^* U} \quad \text{and} \quad k_f = \frac{\delta E}{q^* V}. \quad (\text{A.35})$$

These k_n and k_f are used instead of the raw stocks when solving the model and likelihood. To calculate standard errors, we compute the Hessian on the raw data at the optimum $\hat{\Omega}$ estimated from the scaled data.

B Data

B.1 Occupation Types

This section provides an overview of the individual occupations, as defined by their five digit SOC codes, that are included within each occupation type. Note that the individual occupations are the same as those selected in [Goldin \(2014\)](#).

Business

Business Ops

Buyers and Purchasing Agents, Claims Adjusters, Appraisers, Examiners, and Investigators, Compensation, Benefits, and Job Analysis Specialists, Compliance Officers, Cost Estimators, Human Resources Workers, Logisticians and Project Management Specialists, Management Analysts, Market Research Analysts and Marketing Specialists, Meeting, Convention, and Event Planners, Miscellaneous Business Operations Specialists, Public Relations Specialists

Exec's and GMs

Chief Executives, General and Operations Managers

Financial Ops

Accountants and Auditors, Actuaries, Budget Analysts, Credit Analysts, Credit Counselors and Loan Officers, Financial Analysts and Advisors, Financial Examiners, Miscellaneous Financial Specialists

Lawyers and Judges

Judges, Magistrates, and Other Judicial Workers, Lawyers and Judicial Law Clerks

Sales

Advertising Sales Agents, First-Line Supervisors Non-Retail Sales, Insurance Sales Agents, Miscellaneous Sales Representatives, Services, Sales Engineers, Sales Representatives, Wholesale and Manufacturing, Sales and Related Workers, All Other, Securities, Commodities, and Financial Services Sales Agents

Managers

Administrative Services and Facilities Managers, Advertising and Promotions Managers, Architectural and Engineering Managers, Compensation and Benefits Managers, Computer and Information Systems Managers, Construction Managers, Economists, Emergency Management Directors, Financial Managers, Human Resources Managers, Industrial Production Managers, Marketing and Sales Managers, Medical and Health Services Managers, Miscellaneous Managers, Public Relations and Fundraising Managers, Purchasing Managers, Training and Development Managers, Transportation, Storage, and Distribution Managers

Health

Adv. Clin. Prof.

Nurse Anesthetists, Nurse Practitioners, Physician Assistants, Veterinarians

Clin. Prof.

Audiologists, Dietitians and Nutritionists, Miscellaneous Healthcare Diagnosing or Treating Practitioners, Psychologists, Registered Nurses, Therapists

Medical Practitioners

Chiropractors, Dentists, Optometrists, Pharmacists, Physicians, Podiatrists, Surgeons

Other

Admin & Training

Education and Childcare Administrators, Training and development specialists

Other

Actors, Producers, and Directors, Agents and Business Managers of Artists, Performers, and Athletes, Air Traffic Controllers and Airfield Operations Specialists, Aircraft Pilots and Flight Engineers, Ship and Boat Captains and Operators, Writers and Editors

Post-Sec. Teachers

Postsecondary Teachers

Public Safety

Detectives and Criminal Investigators, First-Line Supervisors of Firefighting and Prevention Workers, First-Line Supervisors of Law Enforcement Workers

Science

Natural Scientists

Agricultural and Food Scientists, Astronomers and Physicists, Atmospheric and Space Scientists, Biological Scientists, Chemists and Materials Scientists, Environmental Scientists and Geoscientists, Miscellaneous Life Scientists, Miscellaneous Mathematical Science Occupations, Miscellaneous Physical Scientists, Natural Sciences Managers, Operations Research Analysts

Social Scientists

Architects, Except Naval, Miscellaneous Social Scientists and Related Workers, Surveyors, Cartographers, and Photogrammetrists, Urban and Regional Planners

Tech

Comp. Programmers

Computer and Information Analysts, Software Developers and Computer Programmers

Cybersec. & Infrastr.

Database and Network Administrators and Architects, Information Security Analysts, Miscellaneous Computer Occupations

Engineers

Aerospace Engineers, Bioengineers and Biomedical Engineers, Chemical Engineers, Civil Engineers, Computer Hardware Engineers, Electrical and Electronics Engineers, Environmental Engineers, Industrial Engineers, Including Health and Safety, Marine Engineers and Naval Architects, Materials Engineers, Mechanical Engineers, Mining, Petroleum, and Geological Engineers, Miscellaneous Engineers

B.2 The Vacancy Data

This appendix provides additional details on how to construct and scale the distributions of job vacancies over occupations for each state and year. We rely on two sources of data, which respectively inform us about the total number of vacancies and the occupational distributions.

1. *Vacancies: Aggregate Levels.* The [Job Openings and Labor Turnover Survey](#) (JOLTS) is collected by the Bureau of Labor Statistics (BLS) in the United States. The survey questions more than 20,000 establishments about their job openings and hires each month. Based on these questions, the BLS estimates the total number of job openings in the United States. For a vacancy to be measured in JOLTS, three conditions need to be satisfied on the last business day of the month: (i) a position exists and has work available, (ii) the job can start within thirty days, and (iii) there is active recruiting for workers from outside of the establishment. The BLS also reports the number of job openings *per unemployed worker*, which allows for simple scaling with our sample. We aggregate their monthly data into yearly estimates for each region.
2. *Vacancies: Occupational Distribution.* The [National Labor Exchange](#) (NLx) Research Hub job vacancy database contains information on individual job postings in the United States. These vacancies identify several characteristics of the job. We use information on the timing, region of employment, and occupation to construct the regional distributions of occupational vacancies. We discuss more features of this data in section [B.3](#).

B.3 The National Labor Exchange (NLx) Research Hub Data

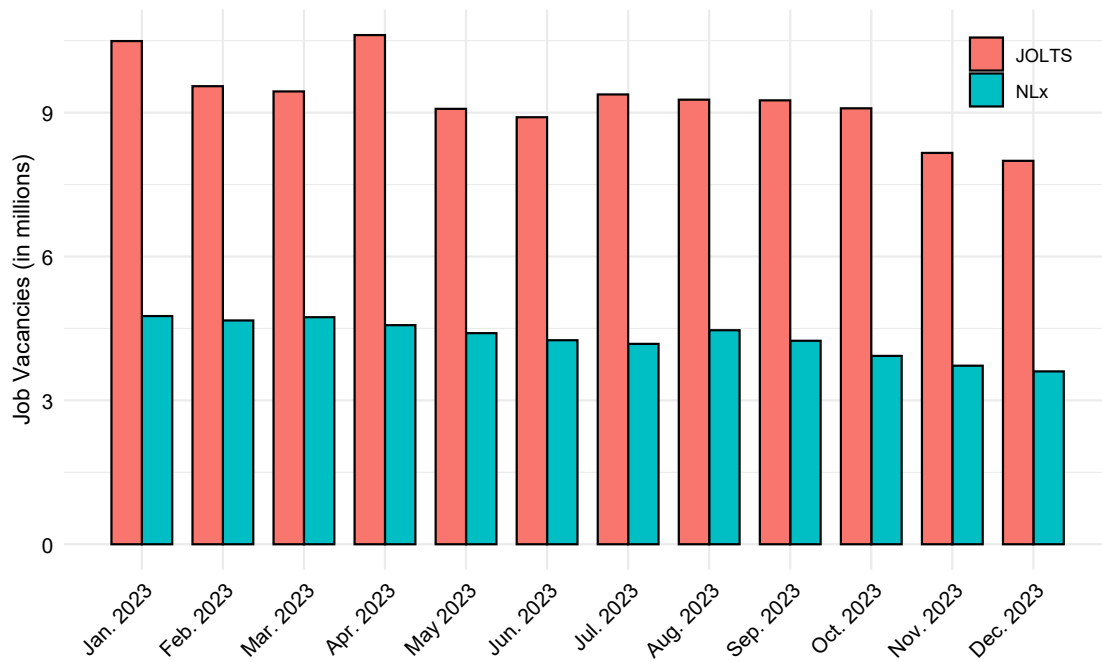
The National Labor Exchange (NLx) is a non-profit online labor exchange operated by the [National Association of State Workforce Agencies](#) (NASWA), which represents the workforce agencies of all states in the United States, and by the [DirectEmployers Association](#) (DE). The exchange aggregates job vacancies from various individual sources, such as companies' career websites, states' job vacancy banks, and the federal jobs portal. The vacancies are vetted to omit junk records and remove duplicates. The Research Hub of the NLx provides data – both real-time and historical snapshots – about all the job vacancies available on the exchange.

The historical snapshots offer information on all open vacancies on any given day, and the NLx provides pooled monthly versions. To construct the job vacancy distribution, we keep all unique vacancies that open in a given month. We then pool all months to estimate the *distribution* of job vacancies at the state $\times$ occupation level. Vacancies which are open for several months are counted multiple times as in JOLTS. While we could filter these out, including them likely yields a more accurate representation of the vacancies that workers face a given moment within the year. We scale these distributions by the number of job vacancies per unemployed worker from JOLTS and, multiplied with the scaled number of unemployed workers in our ACS sample.

Coverage. The NLx aims to cover all online job postings in the United States, but not all vacancies are posted online. Previous work by [Hershbein and Kahn \(2018\)](#), [Deming and Noray \(2020\)](#), and [Acemoglu et al. \(2022\)](#) has assessed how representative vacancies are, using a commercial dataset by [lightcast.io](#). They found that online vacancies are generally representative in terms of industries and occupations. The main shortcoming – see for example Section 3.A in [Hershbein and Kahn \(2018\)](#) – is that highly skilled occupations tend to be over represented. But we focus particularly on this segment of the market in our application.

To better understand the total coverage we compare the number of vacancies in the NLx and JOLTS. Figure 9 highlights both monthly numbers and the movement over time within our sample year. The NLx data contains on average 4.29 million vacancies per month, representing about 46.3% of JOLTS vacancies. Note that this is the total number, and that coverage in the subset of highly skilled vacancies will be even higher. The correlation between both datasets over time ($\rho = 0.83$) is also large.

Figure 9: Vacancies in NLx and JOLTS



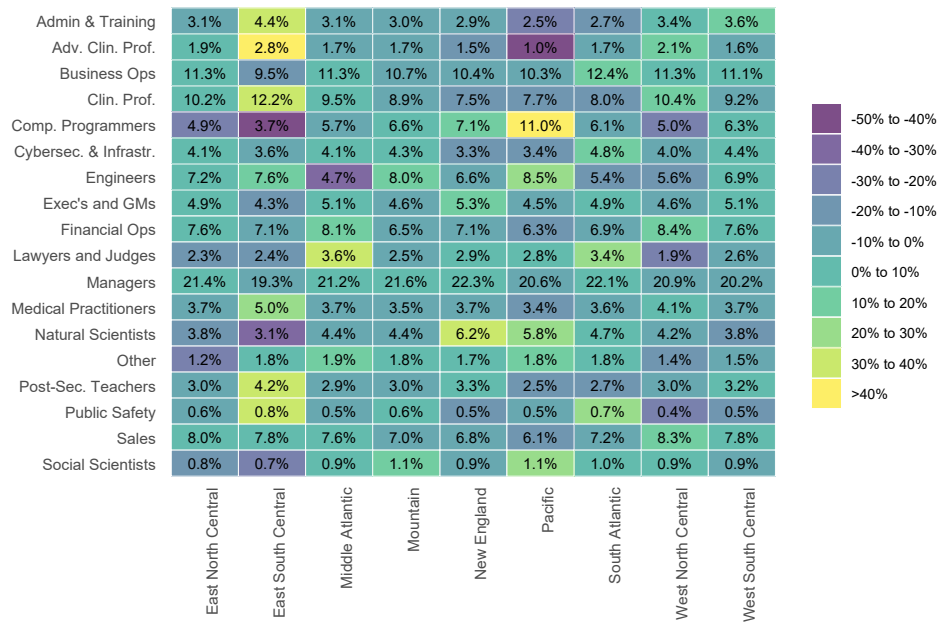
Notes. This figure highlights the monthly number of job vacancies reported by [Job Openings and Labor Turnover Survey](#) (JOLTS) and computed individual postings in the [National Labor Exchange](#) (NLx) Research Hub data.

B.4 Variation in Marginal Distributions Across Markets

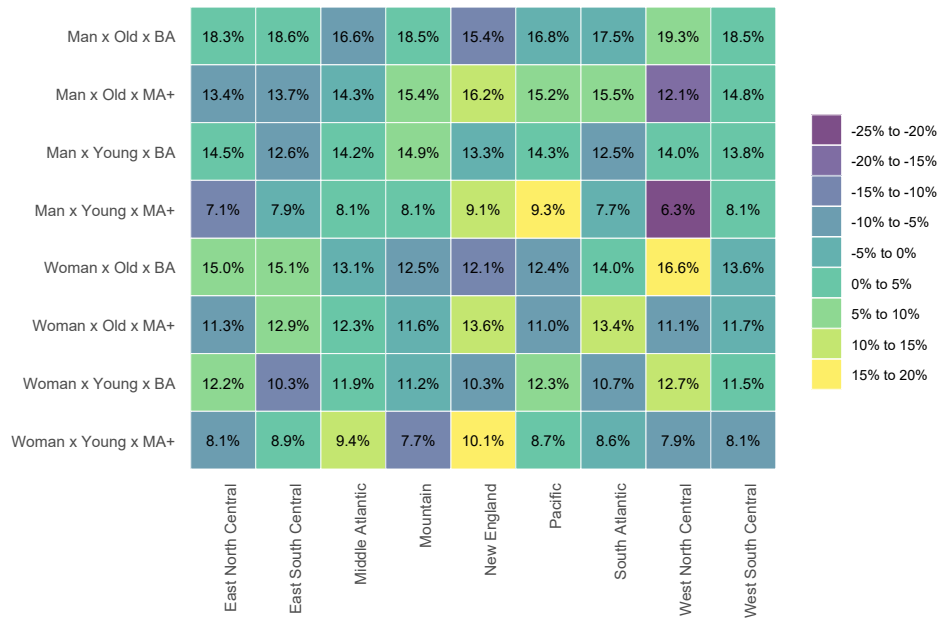
To identify the matching function elasticity, we rely on variation in the marginal distributions of workers and firms across markets. Figure 10 presents these distributions. Panel (a) highlights the differences in worker types. While these are overall relatively similar – as expected – there are some notable differences. For example, Administration and Training occupations represent about 4.4% of the market in the East South Central region, compared to only 2.5% in the Pacific region. The largest difference is for Computer Programmers, who make up 11% in the Pacific region but just 3.7% of the East South Central region. We see the smallest relative variation for Managers, who make up 19.3% of the East South Central region and 22.3% in New England. Panel (b) highlights the comparable, but slightly larger, differences in worker types. For example, young men with at least a Master’s degree account for 9.3% of the Pacific market but just 6.3% in West North Central. Similarly, older men with at least a Master’s degree account for 16.2% of observations in New England, compared just 12.1% in the West North Central region. We conclude that there is sufficient variation in the marginals across these different markets.

Figure 10: Workers and Occupation across Markets

(a) Occupation Types



(b) Workers Types



Notes. This figure shows the number of workers and occupations of each type across the Census Divisions in the United States (2023). The numbers in each cell indicate the fraction of observations within the Census Division that this cell represents. Colors indicate relative deviations from the occupational mean across census divisions. The data is a combination of the [American Community Survey \(ACS\)](#), the [Job Openings and Labor Turnover Survey \(JOLTS\)](#) the [National Labor Exchange \(NLx\)](#) Research Hub job vacancy data.

B.5 Summary Statistics

Table 4 presents further summary statistics about the estimation sample, which is constructed from the 2023 wave of the [American Community Survey](#) (ACS). The sample consists of almost 275,000 individuals and contains slightly more men ($\approx 54\%$) than women, since our focus is on high earning occupations. Because of this sample selection, the unemployment rate is also low ($\approx 2\%$). All individuals in our sample hold a college degree, and 44% have a Master's degree or more, with women being slightly higher educated than men. We selected only individuals ages 25 to 64, and about 58% are 40 years or older, with men being slightly older.

Table 4: Summary Statistics

	Men	Women	Total
Observations			
Employed	148,652	125,892	274,544
Unemployed	2,725	2,519	5,244
Demographics			
Master's Degree +	0.42	0.45	0.44
Age ≥ 40	0.6	0.56	0.58
Occupations			
Business	0.55	0.55	0.55
Health	0.07	0.22	0.14
Other	0.07	0.09	0.08
Science	0.06	0.05	0.06
Tech	0.25	0.08	0.17
Hours			
Full-Time	0.72	0.8	0.76
Long Hours	0.28	0.2	0.24
Wages			
Avg. Month. Wage (in k\$)	12.99	9.59	11.43
Std. Dev. Month. Wage (in k\$)	10.78	7.63	9.62
Avg. Hr. Wage (in \$)	68.12	51.99	60.72
Std. Dev. Hr. Wage (in \$)	53.25	38.23	47.64

Notes. Summary statistics from the latest [American Community Survey](#) (ACS) wave 2023. The selected sample consists of higher educated workers in high earning degrees as introduced in [Goldin \(2014\)](#) and discussed in section 3.

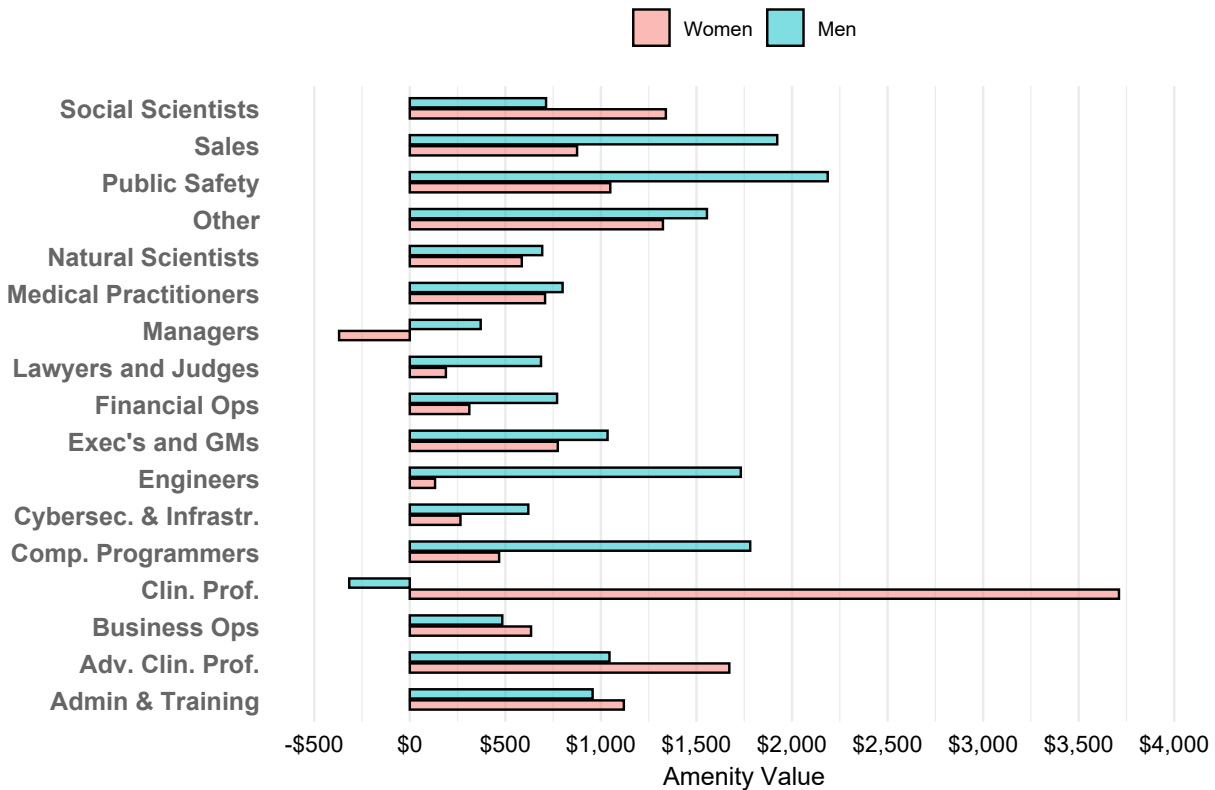
The occupations of men and women are relatively similar, about 55% works in Business, 14% in Health, 17% in Tech, 6% in Science, and 8% in Other occupations. The main difference is that women work more in Health (22% compared to 7%) and men work more in Tech (25% compared

to 8%). We also find that men (28%) are more likely than women (20%) to work long hours. The raw gender gap in monthly earnings is just over 25%, with men earning about 13.000 dollars per month compared to women's 9.500 dollars before taxes. There is also more variance in men's earnings than in women's. The gender gap in hourly wages is just below 25%, with men earning 68 dollars compared to women's 52 dollars, again with higher variance among men.

C Results

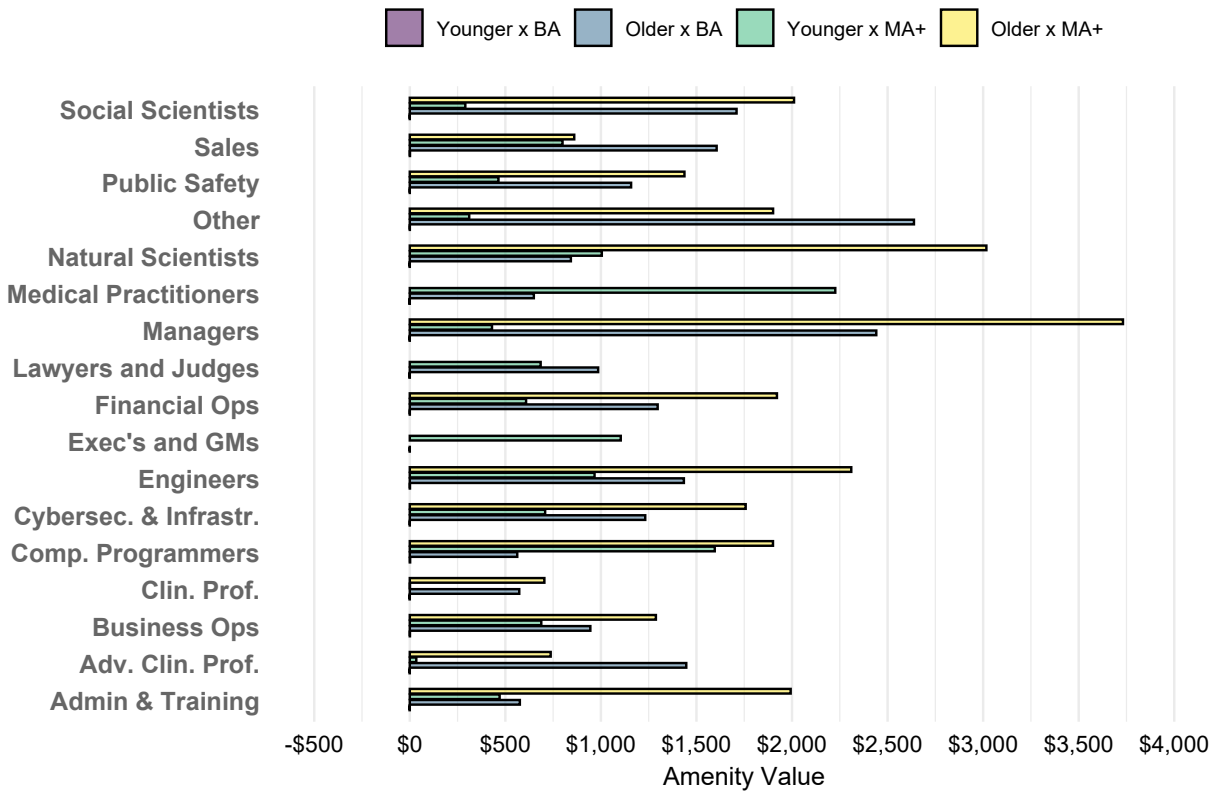
We now present further parameters from the structural model that were omitted from the main text. We first show additional worker-side amenity parameters, then the firm-side productivity values, and finally present the other remaining estimates.

Figure 11: Amenity Value by Gender across Occupations



Notes. This figure shows the estimated amenity value across occupations. The value of Post-Secondary Teachers is normalized to zero. All values are expressed in dollars of monthly earnings.

Figure 12: Productivity Values by Occupations across Workers



Notes. This figure shows the estimated productivity premiums across occupations. The value of Post-Secondary Teachers is normalized to zero. All values are expressed in dollars of monthly earnings.